# Gravity-Constrained Post-Training of JEPA World Models: Convergence and Test-Time Guarantees

## 1 Preliminaries

**Notation.** $\|\cdot\|$ denotes the Euclidean norm and $\langle\cdot,\cdot\rangle$ the Euclidean inner product. All gradients are taken with respect to the predictor parameters $\theta \in \mathbb{R}^D$. For a finite index set $\mathcal{I}$, $|\mathcal{I}|$ denotes its cardinality. $\mathcal{U}(\mathcal{G})$ denotes the uniform distribution on a set $\mathcal{G}$.

**Setting.** Each video shows $K$ objects moving under a uniform gravitational acceleration $g \in \mathbb{R}^2$, which is constant within an episode and varies across episodes. We write $x_t$ for the frame at step $t$ and $\Delta t$ for the frame interval. Every window used below ends before the first contact, so that gravity is the only force acting on the objects within the window. A *real window* is a pair $w = (x_{\le t+H}, g)$ consisting of the frames up to step $t+H$ and the gravity of its episode.

**JEPA world model.** We consider a joint-embedding predictive architecture (JEPA) [1]. The encoder $E$ is pretrained and frozen, and $z_t = E(x_t)$. The predictor $P_\theta$ maps a context $z_{\le t}$ and a conditioning gravity to an autoregressive rollout $\hat{z}_{t+1:t+H} = P_\theta(z_{\le t}, g)$ of horizon $H \ge 2$. The JEPA loss is

$$L(\theta) = \mathbb{E}\Big[\sum_{s=1}^{H} \big\|\hat{z}_{t+s} - z_{t+s}\big\|^2\Big], \tag{1}$$

where the expectation is over real windows and each rollout is conditioned on the window's own gravity $g$. Since the targets $z_{t+s}$ are produced by the frozen encoder, they carry no gradient. Note that $L(\theta) \ge 0$.

**Free fall in discrete time.** The only physical law we use is the following identity.

**Lemma 1.** *Let an object have position $p_t$ and velocity $v$ at step $t$, and move freely under gravity $g$ before $t$ and $g'$ after $t$, so that for every $s \ge 0$,*

$$p_{t+s} = p_t + v\, s\Delta t + \tfrac{1}{2} g'(s\Delta t)^2, \qquad p_{t-s} = p_t - v\, s\Delta t + \tfrac{1}{2} g(s\Delta t)^2.$$

*Then the finite-difference acceleration $a_\tau = (p_{\tau+1} - 2p_\tau + p_{\tau-1})/\Delta t^2$ satisfies $a_t = \frac{1}{2}(g + g')$ and $a_\tau = g'$ for $\tau \ge t+1$, independently of $v$.*

**Position probe.** The probe $h$ is trained on top of the frozen encoder to regress the positions of the $K$ objects and is then frozen. For a latent at step $\tau$, whether encoded from a real frame or predicted, $p_{k,\tau} \in \mathbb{R}^2$ denotes the position of object $k$ decoded by $h$. We write $p^\star_{k,\tau}$ for the true position and

$$e_{k,\tau} := p_{k,\tau} - p^\star_{k,\tau}$$

for the decoding error. On latents encoded from real frames, we assume that the probe has root-mean-square error at most $\delta$:

$$\mathbb{E}\|e_{k,\tau}\|^2 \le \delta^2 \qquad \text{for every object } k \text{ and every frame of a real window.} \tag{2}$$

**Counterfactual constraint.** Given a real window $w$, we sample a counterfactual gravity $g' \sim \mathcal{U}(\mathcal{G})$, where $\mathcal{G}$ is wider than the range of training gravities, and roll the predictor out under it, $\hat{z}_{t+1:t+H} = P_\theta(z_{\le t}, g')$. The positions $p_{k,t-1}, p_{k,t}$ are decoded from the real latents $z_{t-1}, z_t$, and $p_{k,t+s}$, $s = 1, \dots, H$, from the predicted latents. With

$$a_{k,\tau} = \frac{p_{k,\tau+1} - 2p_{k,\tau} + p_{k,\tau-1}}{\Delta t^2}, \qquad \bar{g}_t = \tfrac{1}{2}(g + g'), \qquad \bar{g}_\tau = g' \quad (\tau > t),$$

the free-fall violation of one rollout and the constraint function are

$$c = \frac{1}{KH} \sum_{k=1}^{K} \sum_{\tau=t}^{t+H-1} \left\| a_{k,\tau} - \bar{g}_\tau \right\|^2, \qquad C(\theta) = \mathbb{E}_{\mathbb{P}}[c], \tag{3}$$

where $\mathbb{P}$ denotes the joint distribution of a real window $w$ and an independent $g' \sim \mathcal{U}(\mathcal{G})$. By Lemma 1, $\bar{g}_\tau$ is exactly the finite-difference acceleration of the true counterfactual trajectory. Hence $c = 0$ if and only if the decoded predictions coincide with the free-fall trajectory under $g'$ that starts from the decoded context positions $p_{k,t-1}, p_{k,t}$, i.e., $e_{k,t+s} = (1+s)e_{k,t} - s\, e_{k,t-1}$ for all $k$ and $s$ (Lemma 7 in Appendix D).[1]

**Tolerance.** Let $c_{\text{real}}$ denote $c$ evaluated on held-out real windows with the real future latents $z_{t+1:t+H}$ in place of the predictions and with $g' = g$. For a relative margin $\rho > 0$, we set

$$\varepsilon = (1 + \rho)\, \mathbb{E}[c_{\text{real}}]. \tag{4}$$

We assume $\mathbb{E}[c_{\text{real}}] > 0$, so that $\varepsilon > 0$. Through (4), the probe accuracy $\delta$ determines the scale of the tolerance (Lemma 2); this is the only way in which $\delta$ enters the convergence analysis.

**Lemma 2.** *Under* (2), $\mathbb{E}[c_{\text{real}}] \le 16\delta^2/\Delta t^4$. *Consequently,* $\varepsilon \le 16(1+\rho)\delta^2/\Delta t^4$.

**Problem.** Post-training solves

$$\min_\theta \; L(\theta) \qquad \text{subject to} \qquad C(\theta) \le \varepsilon. \tag{5}$$

The JEPA loss requires future frames and is therefore informative only about the gravities present in the data. The constraint requires no future frames and is the only training signal about gravities outside the training range.

## 2 Algorithm

At iteration $n$, a minibatch $B_n$ of real windows is drawn, each window with its own $g' \sim \mathcal{U}(\mathcal{G})$. Let $L_{B_n}$ and $C_{B_n}$ denote the averages over $B_n$ of the integrand in (1) and of $c$ in (3), respectively. We write

$$\widehat{\nabla} L_n := \nabla L_{B_n}(\theta_n), \qquad \widehat{\nabla} C_n := \nabla C_{B_n}(\theta_n)$$

**Algorithm 1** Gravity-constrained post-training (analyzed form)

**Require:** initial predictor $\theta_0$; step size $\eta$; number of iterations $N$; tolerance $\varepsilon > 0$; projection gate $\gamma \in (0, 1)$

1: **for** $n = 0, 1, \ldots, N-1$ **do**
2: sample a minibatch $B_n$ of real windows and $g' \sim \mathcal{U}(\mathcal{G})$ for each window; compute $\widehat{\nabla} L_n$, $\widehat{\nabla} C_n$, $\widehat{C}_n$
3: **if** $\widehat{C}_n > \varepsilon$ **then**
4: $d_n \leftarrow \widehat{\nabla} C_n / \varepsilon$ ▷ RESTORE
5: **else**
6: $d_n \leftarrow \widehat{\nabla} L_n$ ▷ OPTIMISE
7: **if** $\widehat{C}_n > \gamma\varepsilon$ **and** $\langle \widehat{\nabla} C_n, \widehat{\nabla} L_n \rangle < 0$ **then**
8: $d_n \leftarrow \widehat{\nabla} L_n - \dfrac{\langle \widehat{\nabla} C_n, \widehat{\nabla} L_n \rangle}{\|\widehat{\nabla} C_n\|^2} \widehat{\nabla} C_n$
9: **end if**
10: **end if**
11: $\theta_{n+1} \leftarrow \theta_n - \eta\, d_n$
12: **end for**
13: **return** $\theta_{n^*}$ with $n^* \in \arg\min_{n \in \mathcal{I}_O} \|d_n\|$ (return $\theta_N$ if $\mathcal{I}_O = \emptyset$)

for the minibatch gradients of the JEPA loss and of the constraint, and $\widehat{C}_n$ for an estimate of $C(\theta_n)$, such as $C_{B_n}(\theta_n)$ (Remark 13 discusses an exponential moving average).

We denote by

$$\mathcal{I}_O := \{0 \le n < N : \ \widehat{C}_n \le \varepsilon\}, \qquad \mathcal{I}_R := \{0 \le n < N : \ \widehat{C}_n > \varepsilon\}$$

the OPTIMISE and RESTORE iterations. An OPTIMISE iteration is called *projected* if the projection in line 8 is applied. The projection removes the component of $\widehat{\nabla} L_n$ that conflicts with $\widehat{\nabla} C_n$ [5], so that, to first order, a projected step leaves the minibatch constraint unchanged.

*Remark* 1 (Relation to the practical algorithm). Algorithm 1 differs from the practical version in four respects. First, it switches with a single threshold $\varepsilon$ instead of the hysteresis pair $(\varepsilon, 0.8\varepsilon)$. Hysteresis is not needed for the analysis, since the number of RESTORE iterations is controlled by a potential argument (Lemma 6), and in our analysis it does not improve the resulting bound. Moreover, with $\rho = 0.25$ in (4), the exit threshold $0.8\varepsilon$ equals $\mathbb{E}[c_{\text{real}}]$. Since projected steps keep $C$ approximately constant, the iterates can remain at the exit level, i.e., at the real-data violation level, and every RESTORE phase must push $C$ below that level before it can exit. Second, the switching rule uses a generic estimate $\widehat{C}_n$; the exponential moving average is one choice, analyzed in Remark 13. Third, the projection gate is written as $\gamma\varepsilon$. We recommend $(\gamma - \zeta_C)(1 + \rho) > 1$, where $\zeta_C$ is the relative accuracy of $\widehat{C}_n$ (Assumption 2), e.g., $\rho = 0.25$, $\gamma = 0.9$ and $\zeta_C \le 0.05$. Then $(\gamma - \zeta_C)\varepsilon > \mathbb{E}[c_{\text{real}}]$, so the constraint qualification (Assumption 3) is only required above the real-data violation level. Fourth, the violation in (3) is squared, whereas the method note uses the unsquared norm; hence $c$, $\mathbb{E}[c_{\text{real}}]$ and $\varepsilon$ are different quantities in the two versions.

[1] We square the norm in (3). The unsquared version is not differentiable where $a_{k,\tau} = \bar{g}_\tau$, so the smoothness condition in Assumption 1 below need not hold for it.

# 3 Main Results

## 3.1 Assumptions

**Assumption 1** (Smoothness). *There exist $\ell, M > 0$ and a set $\Theta \subset \mathbb{R}^D$ containing every segment $[\theta_n, \theta_{n+1}]$, $0 \le n < N$, such that for all $\theta, \theta' \in \Theta$,*

$$\|\nabla L(\theta) - \nabla L(\theta')\| \le \ell\|\theta - \theta'\|, \qquad \|\nabla L(\theta)\| \le M,$$
$$\|\nabla C(\theta) - \nabla C(\theta')\| \le \varepsilon\ell\|\theta - \theta'\|, \qquad \|\nabla C(\theta)\| \le \varepsilon M.$$

The constants for $C$ are scaled by $\varepsilon$ because the RESTORE direction is $\widehat{\nabla} C_n/\varepsilon$. Assumption 1 holds, for example, when $P_\theta$ and $h$ have bounded and Lipschitz derivatives on a bounded set that contains the trajectory; it generally fails for ReLU networks, and we impose it along the trajectory only.

**Assumption 2** (Estimation accuracy). *There exist $\zeta, \zeta_C \ge 0$ such that for all $0 \le n < N$,*

$$\|\widehat{\nabla} L_n - \nabla L(\theta_n)\| \le \zeta, \qquad \|\widehat{\nabla} C_n - \nabla C(\theta_n)\| \le \varepsilon\zeta, \qquad |\widehat{C}_n - C(\theta_n)| \le \varepsilon\zeta_C.$$

Since the estimates are random, Assumption 2 is an event; all conclusions below hold on this event. Corollary 1 gives batch sizes under which it holds with high probability.

**Assumption 3** (Constraint qualification). *There exists $\mu \in (0, M]$ such that $\|\nabla C(\theta)\| \ge \varepsilon\mu$ for every $\theta \in \Theta$ with $C(\theta) \ge (\gamma - \zeta_C)\varepsilon$.*

Assumption 3 excludes stationary points of $C$ at violation levels at or above $(\gamma - \zeta_C)\varepsilon$, i.e., infeasible stationary points at which RESTORE would stall. It plays the role of the uniform Slater condition used for switching subgradient methods with weakly convex constraints [4]. Moreover, $\mu \le M$ whenever some $\theta \in \Theta$ satisfies $C(\theta) \ge (\gamma - \zeta_C)\varepsilon$, since then $\varepsilon\mu \le \|\nabla C(\theta)\| \le \varepsilon M$; otherwise Assumption 3 is vacuous and we take $\mu = M$.

## 3.2 Approximate KKT points

We measure the quality of a parameter by the residuals of the Karush–Kuhn–Tucker (KKT) conditions of (5).

**Definition 1** (Approximate KKT point). Let $r_{\mathrm{S}}, r_{\mathrm{F}}, r_{\mathrm{C}} \ge 0$. A parameter $\theta$ is an $(r_{\mathrm{S}}, r_{\mathrm{F}}, r_{\mathrm{C}})$-KKT point of (5) if there exists a multiplier $\nu \ge 0$ such that

$$\|\nabla L(\theta) + \nu\nabla C(\theta)\| \le r_{\mathrm{S}}, \qquad C(\theta) \le (1 + r_{\mathrm{F}})\,\varepsilon, \qquad \big|\nu\big(\varepsilon - C(\theta)\big)\big| \le r_{\mathrm{C}}. \tag{6}$$

The three inequalities in (6) are approximate stationarity of the Lagrangian, primal feasibility, and complementary slackness, respectively; dual feasibility $\nu \ge 0$ is part of the definition. A $(0, 0, 0)$-KKT point satisfies the KKT conditions of (5) exactly. The feasibility residual $r_{\mathrm{F}}$ is measured relative to the tolerance $\varepsilon$.

**Theorem 1.** *Suppose Assumptions 1 to 3 hold and*

$$\zeta \le \frac{\mu^2}{16M}, \qquad \ell\eta \le \frac{\mu^2}{48M^2}, \qquad 2\zeta_C + 6\eta M^2 \le 1 - \gamma, \qquad \zeta_C < \gamma. \tag{7}$$

*Then the iterates of Algorithm 1 satisfy the following.*

*(a)* Primal feasibility. *$C(\theta_n) \le (1+\zeta_C)\varepsilon$ for every $n \in \mathcal{I}_{\mathrm{O}}$. If $N > 4C(\theta_0)/(\varepsilon\eta\mu^2)$, then $\mathcal{I}_{\mathrm{O}} \neq \emptyset$, its first element $n_0$ satisfies $n_0 < 1 + 4C(\theta_0)/(\varepsilon\eta\mu^2)$, and*

$$C(\theta_n) \le \big(1 + \zeta_C + 2\eta M(\zeta + \ell\eta M)\big)\varepsilon \qquad \text{for all } n_0 \le n \le N.$$

*(b)* Stationarity. *The number of OPTIMISE iterations satisfies*

$$|\mathcal{I}_{\mathrm{O}}| \ge \frac{3}{5}\Big(N - \frac{4C(\theta_0)}{\varepsilon\eta\mu^2}\Big),$$

*and, whenever $\mathcal{I}_{\mathrm{O}} \neq \emptyset$,*

$$\frac{1}{|\mathcal{I}_{\mathrm{O}}|}\sum_{n\in\mathcal{I}_{\mathrm{O}}} \|d_n\|^2 \le \frac{4}{\eta|\mathcal{I}_{\mathrm{O}}|}\Big(L(\theta_0) + \frac{12M^2}{\mu^2}\cdot\frac{C(\theta_0)}{\varepsilon}\Big) + 4\zeta^2\Big(1 + \frac{288M^4}{\mu^4}\Big). \tag{8}$$

*(c)* Dual feasibility and complementary slackness. *For every $n \in \mathcal{I}_{\mathrm{O}}$ there exists $\nu_n \ge 0$ such that*

$$\|\nabla L(\theta_n) + \nu_n \nabla C(\theta_n)\| \le \|d_n\| + \frac{5M\zeta}{\mu}, \qquad \varepsilon\nu_n \le \frac{4M}{\mu}, \qquad \big|\nu_n\big(\varepsilon - C(\theta_n)\big)\big| \le \frac{4M}{\mu}\,(1-\gamma+\zeta_C). \tag{9}$$

*Consequently, whenever $\mathcal{I}_{\mathrm{O}} \neq \emptyset$, the output $\theta_{n^*}$ is an $(r_{\mathrm{S}}, \zeta_C, r_{\mathrm{C}})$-KKT point of (5), where*

$$r_{\mathrm{S}} = \Big(\frac{4}{\eta|\mathcal{I}_{\mathrm{O}}|}\Big(L(\theta_0) + \frac{12M^2}{\mu^2}\cdot\frac{C(\theta_0)}{\varepsilon}\Big) + 4\zeta^2\Big(1 + \frac{288M^4}{\mu^4}\Big)\Big)^{1/2} + \frac{5M\zeta}{\mu}, \qquad r_{\mathrm{C}} = \frac{4M}{\mu}\,(1-\gamma+\zeta_C). \tag{10}$$

The proof is given in Appendix B.

*Remark* 2 (The three residuals). The residuals in (10) are controlled by different quantities. The stationarity residual $r_{\mathrm{S}}$ decreases as $N$ grows, up to a floor proportional to $\zeta$ (Remark 4). The feasibility residual equals the relative accuracy $\zeta_C$ of $\widehat{C}_n$; with exact constraint values, every OPTIMISE iterate is feasible, and after the initial RESTORE phase the violation never exceeds $\varepsilon$ by more than $\zeta_C\varepsilon + O(\eta\zeta + \eta^2)\varepsilon$ (part (a)). The complementarity residual $r_{\mathrm{C}}$ is determined by the gate $\gamma$ and by $\zeta_C$, and it does not decrease with $N$. Its dependence on $1-\gamma$ cannot be removed for a fixed gate: projected steps keep $C$ approximately constant, so an iterate that enters the projected regime at a level close to $\gamma\varepsilon$ can remain there with a positive multiplier. Proposition 1 in Appendix B makes this precise: in a one-dimensional example with exact gradients and constraint values, the output satisfies stationarity and feasibility exactly, but its complementarity residual is at least $(1-\gamma)/7$ for all sufficiently large $N$. Making all three residuals small therefore requires $\gamma \to 1$; since (7) requires $6\eta M^2 \le 1-\gamma$, the step size must decrease accordingly (Corollary 1).

*Remark* 3 (Role of the projection). On a projected iteration, $d_n = \widehat{\nabla}L_n + \nu_n\widehat{\nabla}C_n$ with $\nu_n = -\langle\widehat{\nabla}C_n, \widehat{\nabla}L_n\rangle/\|\widehat{\nabla}C_n\|^2 > 0$, and on an unprojected one $d_n = \widehat{\nabla}L_n$. Hence $\|d_n\|$ is exactly the norm of the minibatch Lagrangian gradient with a nonnegative multiplier, which is why the stationarity residual in (c) is controlled by (b). Without the projection, $\|\nabla L(\theta_n)\|$ need not vanish on OPTIMISE iterations near an active constraint, and a different stationarity measure is required, as in the analysis of switching subgradient methods [4].

*Remark* 4 (Rate and noise floor). With exact gradients ($\zeta = 0$), the right-hand side of (8) is $O(1/(\eta N))$, so the Lagrangian residual of the output is $O((\eta N)^{-1/2})$ with a constant step size. With minibatch gradients, the term $4\zeta^2(1 + 288M^4/\mu^4)$ does not vanish as $\eta \to 0$ or $N \to \infty$. The floor reflects the worst-case error model of Assumption 2 and the use of the minibatch quantity $\|d_n\|$ as the certificate. The projection need not preserve unbiasedness of the minibatch gradients, and the proof does not rely on cancellation of the estimation errors. Corollary 1 removes the floor by taking batch sizes inversely proportional to the square of the target accuracy.

*Remark* 5 (Comparison with switching methods). The switching (sub)gradient method was proposed by [2]. Its stochastic version [3] switches on a sample estimate of the constraint and attains $O(1/\sqrt{N})$ optimality gap and constraint violation for convex problems. For weakly convex objectives, [4] establish $O(\epsilon^{-4})$ complexity to a nearly $\epsilon$-stationary point; their stochastic extension, which also switches on sample estimates, covers convex constraints, while their analysis with a weakly convex constraint uses exact oracles, a feasible initialization, and Polyak step sizes on constraint steps. None of these results covers a projected OPTIMISE direction, or a nonconvex constraint whose value is only estimated. Theorem 1 assumes smoothness, a special case of weak convexity, and covers both.

**Corollary 1** (Approximate KKT point)**.** *Suppose Assumptions 1 and 3 hold almost surely, with deterministic $\ell, M, \mu$ and $\Theta$, and let $\mathcal{F}_n$ denote the $\sigma$-algebra generated by $\theta_0, B_0, \dots, B_{n-1}$. Suppose that, conditionally on $\mathcal{F}_n$, the samples in $B_n$ are independent, the per-sample errors of the gradients of $L$ and of $C/\varepsilon$ have zero mean and are norm-subGaussian with parameter $\sigma$ [6], and the per-sample errors of $c/\varepsilon$ have zero mean and are norm-subGaussian (in dimension one) with parameter $\sigma_C$. Let $\widehat{C}_n = C_{B_n}(\theta_n)$. For a target accuracy $r \in (0, 1]$ and a failure probability $q \in (0, 1)$, run Algorithm 1 with*

$$\begin{aligned} \gamma = 1 - \frac{r\mu}{8M}, \qquad \eta = \min\Big\{\frac{\mu^2}{48\ell M^2},\ \frac{r\mu}{96M^3}\Big\}, \\ |B_n| \geq \frac{c_0}{r^2}\max\Big\{\frac{10^4\sigma^2M^4}{\mu^6},\ \frac{1024\,\sigma_C^2M^2}{\mu^2}\Big\}\log\frac{6DN}{q}, \end{aligned} \tag{11}$$

*where $c_0$ is an absolute constant, and*

$$N \geq \max\Big\{\frac{8C(\theta_0)}{\varepsilon\eta\mu^2},\ \frac{320}{3\eta r^2\mu^2}\Big(L(\theta_0) + \frac{12M^2}{\mu^2}\cdot\frac{C(\theta_0)}{\varepsilon}\Big)\Big\}. \tag{12}$$

*Then, with probability at least $1 - q$, the output $\theta_{n^*}$ is an $(r\mu,\, r/32,\, r)$-KKT point of (5).*

The proof is given in Appendix C. With the problem constants, initial objective quantities and a common region $\Theta$ held fixed as $r$ varies, for small $r$ the step size is $\eta = r\mu/(96M^3)$, so Corollary 1 requires $N = O(r^{-3})$ iterations and $\sum_n |B_n| = O\big(r^{-5}\log(DN/q)\big)$ samples. For stationarity alone, a fixed gate and a fixed step size suffice, and $O(r^{-2})$ iterations are enough (Remark 4); the additional factor $r^{-1}$ is the price of complementary slackness, which requires $1 - \gamma = O(r)$ (Remark 2), so that the third condition in (7) forces $\eta = O(r)$. With exact gradients and constraint values ($\sigma = \sigma_C = 0$), the condition on $|B_n|$ is void and the output is an $(r\mu, 0, r)$-KKT point. As in the stochastic switching methods of [3, 4], which estimate constraint values from samples, the per-iteration sample size grows with the target accuracy.

*Remark* 6 (Algorithmic implications). RESTORE prioritizes physical feasibility and can increase the JEPA loss. Early projection is conservative: with a fixed $\gamma < 1$, learning can stop at a strictly feasible point, and additional iterations alone do not remove this effect (Proposition 1). The theorem separates the roles of iterations, estimation accuracy and the projection gate in controlling the three KKT residuals; the condition $2\zeta_C + 6\eta M^2 \leq 1 - \gamma$ couples these controls through the admissible step size. The guarantee applies to the selected iterate $\theta_{n^*}$; convergence of the full parameter sequence and global optimality are not established.

### 3.3 Test-time error under a new gravity

Let $\mathbb{Q}$ be a distribution of pairs $(w, g')$, where $w$ is a window whose context frames follow free fall under the gravity $g$ of its episode on $[t-1, t]$, and $g'$ is the conditioning gravity at test time.

For $s = 1, \dots, H$, $p^\star_{k,t+s}$ denotes the true trajectory when gravity switches from $g$ to $g'$ at step $t$; when $g' = g$, this is the actual future. The predicted positions $p_{k,t+s}$ are decoded by $h$ from $\hat{z}_{t+1:t+H} = P_\theta(z_{\le t}, g')$.

**Corollary 2** (Test-time position error)**.** *Suppose that, under $\mathbb{Q}$, the trajectories $p^\star_{k,t-1:t+H}$ are contact-free, that the joint law of $(x_{\le t}, g, g', (p^\star_{k,t-1}, p^\star_{k,t})_{k=1}^K)$ under $\mathbb{Q}$ is absolutely continuous with respect to its law under $\mathbb{P}$ with density at most $\kappa$, and that (2) holds, where the real windows in (2), (4) and $\mathbb{P}$ have the same distribution. Let $\theta$ be independent of the test data and satisfy $C(\theta) \le (1+\zeta_C)\varepsilon$; by Theorem 1(a), this holds for $\theta = \theta_{n^*}$ under the hypotheses of Theorem 1 whenever $\mathcal{I}_{\mathrm{O}} \neq \emptyset$. All test expectations below are conditional on $\theta$. Then, for $s = 1, \dots, H$,*

$$\begin{aligned}\left(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\left\|p_{k,t+s} - p^\star_{k,t+s}\right\|^2\right)^{1/2} &\le \sqrt{\kappa}\left[(2s+1)\delta + \Delta t^2\sqrt{\frac{Hs(s+1)(2s+1)}{6}\,C(\theta)}\right] \\ &\le \sqrt{\kappa}\,\delta\left[2s+1+4\sqrt{\frac{(1+\zeta_C)(1+\rho)Hs(s+1)(2s+1)}{6}}\right].\end{aligned} \tag{13}$$

The proof is given in Appendix D. By the Cauchy–Schwarz inequality, (13) also bounds the expected mean position error.

*Remark* 7 (Long-horizon position error). At $s = H$, the first bound in (13) becomes

$$\left(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\left\|p_{k,t+H} - p^\star_{k,t+H}\right\|^2\right)^{1/2} \le \sqrt{\kappa}\left[(2H+1)\delta + \Delta t^2 H\sqrt{\frac{(H+1)(2H+1)}{6}\,C(\theta)}\right].$$

Thus initial probe errors are amplified linearly, whereas persistent acceleration residuals can produce quadratic position drift by (33). With $\kappa, \rho, \zeta_C$ fixed, the calibrated bound is $O(H^2\delta)$; it does not establish an error bound uniform in $H$. The guarantee covers only the contact-free rollout of length $H$ on which the residual is controlled, and does not extend automatically to longer rollouts or to latent and frame prediction errors. The optimization constants in Assumptions 1 and 3 can also depend on $H$.

*Remark* 8 (Scope of Corollary 2).

(i) *Probe coordinates.* The bound concerns positions decoded by the probe $h$ used in training. For a held-out probe $h'$ with the same output coordinates, the additional term $(K^{-1}\mathbb{E}_{\mathbb{Q}}\|h'(\hat{z}_{t+s}) - h(\hat{z}_{t+s})\|^2)^{1/2}$ appears by the $L^2$ triangle inequality; it requires no labels and can be measured at test time.

(ii) *A single test gravity.* If $\mathcal{U}(\mathcal{G})$ is atomless and $g'$ is fixed, $\mathbb{Q}$ is not absolutely continuous with respect to $\mathbb{P}$. By Markov's inequality, however, the set of $g' \in \mathcal{G}$ for which $\mathbb{E}_{\mathbb{P}}[c \mid g'] > \alpha\, C(\theta)$ has $\mathcal{U}(\mathcal{G})$-measure at most $1/\alpha$, for every $\alpha \ge 1$.

(iii) *A new environment.* If the context gravity $g$ lies outside the support of episode gravities under $\mathbb{P}$, the required joint distribution is not covered and no finite $\kappa$ exists. What remains is the pathwise bound of Lemma 7, $\frac{1}{K}\sum_k \|p_{k,t+s} - p^\star_{k,t+s}\| \le sH\Delta t^2\sqrt{c} + \frac{1}{K}\sum_k\big((1+s)\|e_{k,t}\| + s\|e_{k,t-1}\|\big)$, whose first term requires no future frames and can be computed at test time, provided the context gravity $g$, which enters $\bar{g}_t$, is known exactly. If it is estimated, an additional term for its estimation error is required.

(iv) *Latent prediction error.* Neither $L$ nor $C$ constrains the predicted latents in the directions that the probe ignores. Consequently, no bound on $\|\hat{z}_{t+s} - z_{t+s}\|$ under a counterfactual or new gravity can be derived from $L$ and $C$ alone (Proposition 2 in Appendix F).

### 3.4 Comparison with the ablations

Corollary 2 holds for every predictor that satisfies the constraint, and Theorem 1(a) shows that Algorithm 1 produces such a predictor. To quantify the benefit of the method, we compare it with the ablations of the method note: the pretrained predictor, continued JEPA training, and the constraint with $g'$ restricted to the real gravity of each window. We use the counterfactual protocol of the constraint, in which the context is a real window and only the conditioning gravity changes. In this subsection, $\mathbb{Q}$ denotes the law of $(w, g')$, where $w$ is a real window and $g'$ is drawn independently of $w$ from a distribution on $\mathcal{G}$ whose density with respect to $\mathcal{U}(\mathcal{G})$ is at most $\kappa$. For a predictor $\theta$ and a conditioning gravity $g'$, we write $p_{k,t+s}(\theta, g')$ for the position of object $k$ decoded by $h$ from the $s$-step prediction of $P_\theta(z_{\le t}, g')$, and $p^\star_{k,t+s}(g')$ for the true position of object $k$ at step $t+s$ when gravity switches from $g$ to $g'$ at step $t$. In particular, $p^\star_{k,t+s}(g)$ is the actual future of the window.

The ablations are defined as follows. The pretrained predictor $\theta_0$ is called *gravity-blind* if $p_{k,t+s}(\theta_0, g')$ does not depend on $g'$ for every window, object $k$ and step $s$. This is the case when the pretrained predictor takes no gravity input and a FiLM adapter [7] initialized to the identity is added, as in the method note. Continued JEPA training is gradient descent on $L$ with exact gradients,

$$\theta_0^{\mathrm{JEPA}} = \theta_0, \qquad \theta_{n+1}^{\mathrm{JEPA}} = \theta_n^{\mathrm{JEPA}} - \eta \nabla L(\theta_n^{\mathrm{JEPA}}) \qquad (0 \le n < N). \tag{14}$$

The factual constraint is

$$C_{\mathrm{fact}}(\theta) = \mathbb{E}[c], \tag{15}$$

where $c$ is defined in (3) and the expectation is over real windows with $g' = g$. We denote by $\theta_n^{\mathrm{fact}}$ the iterates of Algorithm 1 started at $\theta_0$, with $C_{\mathrm{fact}}$ in place of $C$ and with exact quantities $\widehat{\nabla} L_n = \nabla L(\theta_n^{\mathrm{fact}})$, $\widehat{\nabla} C_n = \nabla C_{\mathrm{fact}}(\theta_n^{\mathrm{fact}})$ and $\widehat{C}_n = C_{\mathrm{fact}}(\theta_n^{\mathrm{fact}})$.

**Theorem 2** (Counterfactual separation)**.** *Suppose that (2) holds, that the trajectories $p^\star_{k,t+1:t+H}(g')$ are contact-free under $\mathbb{Q}$, and let $s \in \{1, \dots, H\}$.*

*(a)* Full method. *Under the hypotheses of Theorem 1, if $\mathcal{I}_{\mathrm{O}} \neq \emptyset$ and $\theta_{n^*}$ is independent of the test data, then*

$$\Big(\frac{1}{K}\sum_{k=1}^{K} \mathbb{E}_{\mathbb{Q}} \big\| p_{k,t+s}(\theta_{n^*}, g') - p^\star_{k,t+s}(g') \big\|^2\Big)^{1/2} \le \sqrt{\kappa}\,\delta\Big[2s + 1 + 4\sqrt{\frac{(1+\zeta_C)(1+\rho)Hs(s+1)(2s+1)}{6}}\Big]. \tag{16}$$

*(b)* Continued JEPA training. *Suppose that $\nabla L$ is $\ell$-Lipschitz on a set $\Theta'$ containing the segments $[\theta_n^{\mathrm{JEPA}}, \theta_{n+1}^{\mathrm{JEPA}}]$, $0 \le n < N$, that $\eta\ell \le 1$, and that there exists $\Lambda \ge 0$ such that, for all $\theta, \theta' \in \Theta'$,*

$$\Big(\frac{1}{K}\sum_{k=1}^{K} \mathbb{E}_{\mathbb{Q}} \big\| p_{k,t+s}(\theta, g') - p_{k,t+s}(\theta', g') \big\|^2\Big)^{1/2} \le \Lambda \|\theta - \theta'\|. \tag{17}$$

*Then, for every $0 \le n \le N$,*

$$\Big(\frac{1}{K}\sum_{k=1}^{K} \mathbb{E}_{\mathbb{Q}} \big\| p_{k,t+s}(\theta_n^{\mathrm{JEPA}}, g') - p^\star_{k,t+s}(g') \big\|^2\Big)^{1/2} \ge \Big(\frac{1}{K}\sum_{k=1}^{K} \mathbb{E}_{\mathbb{Q}} \big\| p_{k,t+s}(\theta_0, g') - p^\star_{k,t+s}(g') \big\|^2\Big)^{1/2} - \Lambda\sqrt{2\eta n\Big(L(\theta_0) - \inf_{\theta \in \mathbb{R}^D} L(\theta)\Big)}. \tag{18}$$

*If, in addition, $\theta_0$ is gravity-blind, then*

$$\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\big\|p_{k,t+s}(\theta_0,g')-p^{\star}_{k,t+s}(g')\big\|^2\Big)^{1/2} \geq \frac{(s\Delta t)^2}{2}\big(\mathbb{E}_{\mathbb{Q}}\|g'-g\|^2\big)^{1/2}-\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}\big\|p_{k,t+s}(\theta_0,g)-p^{\star}_{k,t+s}(g)\big\|^2\Big)^{1/2}, \quad (19)$$

*where the last expectation is over real windows.*

*(c)* Factual constraint. *Suppose, in addition to the hypotheses of (b), that $\|\nabla C_{\mathrm{fact}}(\theta)\|\leq \varepsilon M$ for all $\theta\in\Theta'$ and*

$$C_{\mathrm{fact}}(\theta_0)+\varepsilon M\sqrt{2\eta N\Big(L(\theta_0)-\inf_{\theta\in\mathbb{R}^D}L(\theta)\Big)}\leq\gamma\varepsilon. \quad (20)$$

*Then $\theta_n^{\mathrm{fact}}=\theta_n^{\mathrm{JEPA}}$ for every $0\leq n\leq N$. In particular, the bounds in (b) hold for the iterates and the output of the factual-constraint method.*

The proof is given in Appendix E.

*Remark* 9 (Separation). Suppose $\theta_0$ is gravity-blind. By (18) and (19), the counterfactual error of continued JEPA training after $n\leq N$ steps is at least

$$\frac{(s\Delta t)^2}{2}\big(\mathbb{E}_{\mathbb{Q}}\|g'-g\|^2\big)^{1/2}-\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}\big\|p_{k,t+s}(\theta_0,g)-p^{\star}_{k,t+s}(g)\big\|^2\Big)^{1/2} -\Lambda\sqrt{2\eta n\Big(L(\theta_0)-\inf_{\theta\in\mathbb{R}^D}L(\theta)\Big)}. \quad (21)$$

The first term is the displacement caused by the change of gravity; it grows linearly in the gravity shift and quadratically in $s$. The second term is the factual error of the pretrained predictor in probe coordinates, and the third bounds how far the JEPA loss can move the predictor. The right-hand side of (16), in contrast, does not depend on the gravity shift. Since (21) is nonincreasing in $n$, the full method is strictly more accurate than the pretrained predictor ($n=0$), than every iterate of continued JEPA training, and, under (20), than the iterates and the output of the factual constraint, whenever (21) with $n=N$ exceeds the right-hand side of (16). If $\theta_0$ minimizes $L$, then $\nabla L(\theta_0)=0$, the iterates (14) remain at $\theta_0$, and the last term in (21) vanishes for every $N$. More generally, the smaller the excess loss $L(\theta_0)-\inf_\theta L(\theta)$ of the pretrained predictor, the less the JEPA loss can change its response to counterfactual gravities: the JEPA loss evaluates the predictor only under the gravity of each window, so the counterfactual response changes only as a by-product of fitting real videos.

*Remark* 10 (Roles of counterfactual gravities and of the switching rule). The two ingredients of the method enter Theorem 2 through different hypotheses. Counterfactual gravities make the density bound $\kappa$ finite for conditioning gravities outside the training range. With $g'$ restricted to the gravity of each window, the constraint is evaluated only at factual pairs, and part (c) shows that it then leaves continued JEPA training unchanged. Condition (20) describes the regime in which the pretrained predictor is already physically consistent on real windows: if its decoded predictions are as accurate as decoded real frames, then $C_{\mathrm{fact}}(\theta_0)$ is close to $\mathbb{E}[c_{\mathrm{real}}]=\varepsilon/(1+\rho)$, which is below $\gamma\varepsilon$ under the recommendation $(\gamma-\zeta_C)(1+\rho)>1$ of Remark 1. The switching rule provides the certificate $C(\theta_{n^*})\leq(1+\zeta_C)\varepsilon$, on which (16) relies, without knowledge of the Lagrange multiplier.

A fixed penalty does not provide this certificate. For $D = 1$, $L(\theta) = \frac{1}{2}(\theta - a)^2$ with $a > \sqrt{2}$, and $C(\theta) = \frac{\varepsilon}{2}\theta^2$, the unique stationary point of $L + \lambda C$ satisfies $C \le \varepsilon$ if and only if $\lambda \ge (a/\sqrt{2} - 1)/\varepsilon$, and the right-hand side is the multiplier of (5) at its solution (Appendix E). Within this family, a fixed $\lambda$ therefore certifies feasibility only for problems whose multiplier is at most $\lambda$.

*Remark* 11 (Scope of Theorem 2). The separation concerns the counterfactual protocol, in which the context is a real window. It does not cover a test environment whose context already evolves under a gravity outside the training range (Remark 8(iii)). It also requires the pretrained predictor to respond incorrectly to counterfactual gravities, as a gravity-blind predictor does; if $\theta_0$ already predicts counterfactual futures accurately, the right-hand side of (18) is small and there is nothing to separate. Parts (b) and (c) concern idealized ablations with exact gradients. Finally, all bounds are stated in the coordinates of the training probe $h$, and Remark 8(i) applies.

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## A Auxiliary Results

### A.1 Proof of Lemma 1

For $\tau \ge t + 1$, the positions $p_{\tau-1}, p_\tau, p_{\tau+1}$ are all given by the first formula. Writing $\tau = t + s$ with $s \ge 1$,

$$p_{\tau+1} - 2p_\tau + p_{\tau-1} = v\Delta t\big((s+1) - 2s + (s-1)\big) + \tfrac{1}{2}g'\Delta t^2\big((s+1)^2 - 2s^2 + (s-1)^2\big) = g'\Delta t^2.$$

At $\tau = t$, the first formula with $s = 1$ and the second formula with $s = 1$ give

$$p_{t+1} - 2p_t + p_{t-1} = \big(v\Delta t + \tfrac{1}{2}g'\Delta t^2\big) + \big(-v\Delta t + \tfrac{1}{2}g\Delta t^2\big) = \tfrac{1}{2}(g + g')\Delta t^2.$$

Dividing by $\Delta t^2$ completes the proof. □

### A.2 Proof of Lemma 2

For $c_{\text{real}}$, all positions are decoded from real latents and $g' = g$. Since the window is contact-free, the true positions follow free fall under $g$ on $[t-1, t+H]$, and Lemma 1 with $g' = g$ gives $p^\star_{k,\tau+1} - 2p^\star_{k,\tau} + p^\star_{k,\tau-1} = g\Delta t^2$ for $t \le \tau \le t+H-1$. Hence

$$a_{k,\tau} - \bar{g}_\tau = \frac{e_{k,\tau+1} - 2e_{k,\tau} + e_{k,\tau-1}}{\Delta t^2}.$$

For any vectors $u_1, u_2, u_3$, the Cauchy–Schwarz inequality with weights $(1,2,1)$ gives $\|u_1 + 2u_2 + u_3\|^2 \le 4\big(\|u_1\|^2 + 2\|u_2\|^2 + \|u_3\|^2\big)$. Therefore, for every $k$ and $\tau$,

$$\mathbb{E}\|a_{k,\tau} - \bar{g}_\tau\|^2 \le \frac{4}{\Delta t^4}\Big(\mathbb{E}\|e_{k,\tau+1}\|^2 + 2\,\mathbb{E}\|e_{k,\tau}\|^2 + \mathbb{E}\|e_{k,\tau-1}\|^2\Big) \le \frac{16\,\delta^2}{\Delta t^4},$$

where the last step uses (2). Averaging over $k$ and $\tau$ in (3) yields $\mathbb{E}[c_{\text{real}}] \le 16\delta^2/\Delta t^4$, and (4) yields the bound on $\varepsilon$. □

*Remark* 12. If the decoding errors of different frames are independent with zero mean, the constant 16 improves to 6, since then $\mathbb{E}\|e_{k,\tau+1} - 2e_{k,\tau} + e_{k,\tau-1}\|^2 \le 6\delta^2$.

## B Proof of Theorem 1

Throughout this appendix, Assumptions 1 to 3 and the conditions (7) hold. We use the descent lemma [8, Section 1.2]: if $f$ has an $\ell_f$-Lipschitz gradient on the segment $[\theta, \theta - \eta d]$, then

$$f(\theta - \eta d) \le f(\theta) - \eta\langle \nabla f(\theta), d\rangle + \frac{\ell_f \eta^2}{2}\|d\|^2. \tag{22}$$

We also use two consequences of (7) and $\mu \le M$: $\zeta \le \mu^2/(16M) \le \mu/16$, and $\ell\eta \le 1/48$. In particular, Assumptions 1 and 2 give

$$\|\widehat{\nabla} L_n\| \le M + \zeta \le 2M, \qquad \|\widehat{\nabla} C_n\| \le \varepsilon(M+\zeta) \le 2\varepsilon M \qquad (0 \le n < N). \tag{23}$$

### B.1 Consequences of the switching rule

We call an OPTIMISE iteration *exceptional* if $\langle \widehat{\nabla} C_n, \widehat{\nabla} L_n\rangle < 0$ but the projection is not applied, i.e., if $\langle \widehat{\nabla} C_n, \widehat{\nabla} L_n\rangle < 0$ and $\widehat{C}_n \le \gamma\varepsilon$.

**Lemma 3.** *For every $0 \le n < N$:*

*(i) if $n \in \mathcal{I}_{\text{R}}$, then $C(\theta_n) > (1 - \zeta_C)\varepsilon$;*

*(ii) if $n \in \mathcal{I}_{\text{O}}$, then $C(\theta_n) \le (1 + \zeta_C)\varepsilon$;*

*(iii) if $n \in \mathcal{I}_{\text{O}}$ is projected, then $C(\theta_n) > (\gamma - \zeta_C)\varepsilon$;*

*(iv) if $n \in \mathcal{I}_{\text{O}}$ is exceptional, then $C(\theta_n) \le (\gamma + \zeta_C)\varepsilon$.*

*In particular, since $\gamma < 1$, Assumption 3 applies at $\theta_n$ whenever $n \in \mathcal{I}_{\text{R}}$ or $n$ is projected.*

*Proof.* By Assumption 2, $\widehat{C}_n - \varepsilon\zeta_C \le C(\theta_n) \le \widehat{C}_n + \varepsilon\zeta_C$. Statement (i) follows from $\widehat{C}_n > \varepsilon$ on $\mathcal{I}_{\text{R}}$, (ii) from $\widehat{C}_n \le \varepsilon$ on $\mathcal{I}_{\text{O}}$, (iii) from the gate $\widehat{C}_n > \gamma\varepsilon$ in line 7, and (iv) from $\widehat{C}_n \le \gamma\varepsilon$, which is the only reason a conflicting OPTIMISE iteration is not projected. □

## B.2 The OPTIMISE direction

**Lemma 4.** *For every $n \in \mathcal{I}_{\mathrm{O}}$, the following hold.*

*(i)* $\langle \widehat{\nabla} L_n, d_n \rangle = \|d_n\|^2$ *and* $\|d_n\| \le \|\widehat{\nabla} L_n\|$.

*(ii) If $n$ is not exceptional, then* $\langle \widehat{\nabla} C_n, d_n \rangle \ge 0$.

*(iii)* $d_n = \widehat{\nabla} L_n + \nu_n \widehat{\nabla} C_n$, *where* $\nu_n = -\langle \widehat{\nabla} C_n, \widehat{\nabla} L_n \rangle / \|\widehat{\nabla} C_n\|^2 > 0$ *if $n$ is projected and* $\nu_n = 0$ *otherwise.*

*Proof.* If $n$ is not projected, then $d_n = \widehat{\nabla} L_n$ and $\nu_n = 0$; (i) and (iii) are immediate, and (ii) holds because a non-exceptional, unprojected iteration satisfies $\langle \widehat{\nabla} C_n, \widehat{\nabla} L_n \rangle \ge 0$. If $n$ is projected, then $\langle \widehat{\nabla} C_n, \widehat{\nabla} L_n \rangle < 0$, so $\widehat{\nabla} C_n \neq 0$ and $\nu_n > 0$ is well defined, and $d_n = \widehat{\nabla} L_n + \nu_n \widehat{\nabla} C_n$ by line 8. Then

$$\langle \widehat{\nabla} C_n, d_n \rangle = \langle \widehat{\nabla} C_n, \widehat{\nabla} L_n \rangle + \nu_n \|\widehat{\nabla} C_n\|^2 = 0,$$

which proves (ii). Using this orthogonality,

$$\begin{aligned} \langle \widehat{\nabla} L_n, d_n \rangle &= \langle d_n - \nu_n \widehat{\nabla} C_n, d_n \rangle = \|d_n\|^2, \\ \|\widehat{\nabla} L_n\|^2 &= \|d_n - \nu_n \widehat{\nabla} C_n\|^2 = \|d_n\|^2 + \nu_n^2 \|\widehat{\nabla} C_n\|^2 \ge \|d_n\|^2, \end{aligned}$$

which proves (i). □

## B.3 One-step inequalities

**Lemma 5.** *For every $0 \le n < N$, $\|d_n\| \le 2M$ and the following hold.*

*(i) If $n \in \mathcal{I}_{\mathrm{R}}$, then*

$$C(\theta_{n+1}) \le C(\theta_n) - \frac{\varepsilon \eta \mu^2}{4}, \qquad L(\theta_{n+1}) \le L(\theta_n) + 3\eta M^2. \tag{24}$$

*(ii) If $n \in \mathcal{I}_{\mathrm{O}}$, then*

$$L(\theta_{n+1}) \le L(\theta_n) - \frac{\eta}{2} \|d_n\|^2 + \eta \zeta^2. \tag{25}$$

*Moreover, if $n$ is not exceptional,*

$$C(\theta_{n+1}) \le C(\theta_n) + \varepsilon \eta \Big( \zeta \|d_n\| + \frac{\ell \eta}{2} \|d_n\|^2 \Big), \tag{26}$$

*and if $n$ is exceptional, $C(\theta_{n+1}) \le (\gamma + \zeta_C + 3\eta M^2)\varepsilon$.*

*Proof.* The bound $\|d_n\| \le 2M$ follows from (23), since $d_n = \widehat{\nabla} C_n / \varepsilon$ on $\mathcal{I}_{\mathrm{R}}$ and $\|d_n\| \le \|\widehat{\nabla} L_n\|$ on $\mathcal{I}_{\mathrm{O}}$ by Lemma 4(i).

*RESTORE iterations.* Let $n \in \mathcal{I}_{\mathrm{R}}$, so that $d_n = \widehat{\nabla} C_n / \varepsilon$. By Lemma 3, Assumption 3 applies at $\theta_n$, i.e., $\|\nabla C(\theta_n)\| / \varepsilon \ge \mu$. Applying (22) to $C$, whose gradient is $\varepsilon\ell$-Lipschitz,

$$\begin{aligned} C(\theta_{n+1}) &\overset{(a)}{\le} C(\theta_n) - \frac{\eta}{\varepsilon} \langle \nabla C(\theta_n), \widehat{\nabla} C_n \rangle + \frac{\ell \eta^2}{2\varepsilon} \|\widehat{\nabla} C_n\|^2 \\ &\overset{(b)}{=} C(\theta_n) - \frac{\eta}{\varepsilon} \|\nabla C(\theta_n)\|^2 - \frac{\eta}{\varepsilon} \langle \nabla C(\theta_n), \widehat{\nabla} C_n - \nabla C(\theta_n) \rangle + \frac{\ell \eta^2}{2\varepsilon} \|\widehat{\nabla} C_n\|^2 \\ &\overset{(c)}{\le} C(\theta_n) - \varepsilon \eta \Big( \frac{\|\nabla C(\theta_n)\|^2}{\varepsilon^2} - \zeta \frac{\|\nabla C(\theta_n)\|}{\varepsilon} \Big) + 2\varepsilon \ell \eta^2 M^2 \\ &\overset{(d)}{\le} C(\theta_n) - \varepsilon \eta \big( \mu^2 - \zeta \mu \big) + 2\varepsilon \ell \eta^2 M^2 \\ &\overset{(e)}{\le} C(\theta_n) - \varepsilon \eta \Big( \frac{\mu^2}{2} - \frac{\mu^2}{24} \Big) \le C(\theta_n) - \frac{\varepsilon \eta \mu^2}{4}, \end{aligned}$$

where (a) is (22) with $d = \widehat{\nabla} C_n/\varepsilon$; (b) adds and subtracts $\nabla C(\theta_n)$; (c) uses the Cauchy–Schwarz inequality, Assumption 2, and (23); (d) holds because $x \mapsto x^2 - \zeta x$ is nondecreasing on $[\zeta/2, \infty)$ and $\|\nabla C(\theta_n)\|/\varepsilon \geq \mu \geq \zeta/2$; and (e) uses $\zeta\mu \leq \mu^3/(16M) \leq \mu^2/16 \leq \mu^2/2$ and $2\ell\eta M^2 \leq \mu^2/24$, both from (7). For the JEPA loss,

$$L(\theta_{n+1}) \overset{(a)}{\leq} L(\theta_n) - \frac{\eta}{\varepsilon}\langle \nabla L(\theta_n), \widehat{\nabla} C_n\rangle + \frac{\ell\eta^2}{2\varepsilon^2}\|\widehat{\nabla} C_n\|^2 \overset{(b)}{\leq} L(\theta_n) + 2\eta M^2 + 2\ell\eta^2 M^2 \overset{(c)}{\leq} L(\theta_n) + 3\eta M^2,$$

where (a) is (22); (b) uses the Cauchy–Schwarz inequality, $\|\nabla L(\theta_n)\| \leq M$, and (23); and (c) uses $\ell\eta \leq 1/2$.

*OPTIMISE iterations.* Let $n \in \mathcal{I}_{\mathrm{O}}$. Applying (22) to $L$,

$$\begin{aligned} L(\theta_{n+1}) &\overset{(a)}{\leq} L(\theta_n) - \eta\langle \widehat{\nabla} L_n, d_n\rangle + \eta\langle \widehat{\nabla} L_n - \nabla L(\theta_n), d_n\rangle + \frac{\ell\eta^2}{2}\|d_n\|^2 \\ &\overset{(b)}{\leq} L(\theta_n) - \eta\|d_n\|^2 + \eta\zeta\|d_n\| + \frac{\ell\eta^2}{2}\|d_n\|^2 \\ &\overset{(c)}{\leq} L(\theta_n) - \eta\|d_n\|^2 + \Big(\frac{\eta}{4}\|d_n\|^2 + \eta\zeta^2\Big) + \frac{\eta}{4}\|d_n\|^2 = L(\theta_n) - \frac{\eta}{2}\|d_n\|^2 + \eta\zeta^2, \end{aligned}$$

where (a) is (22) after adding and subtracting $\widehat{\nabla} L_n$; (b) uses Lemma 4(i), the Cauchy–Schwarz inequality, and Assumption 2; and (c) uses $\zeta\|d_n\| \leq \frac{1}{4}\|d_n\|^2 + \zeta^2$ and $\ell\eta \leq \frac{1}{2}$. This proves (25).

If $n$ is not exceptional, applying (22) to $C$,

$$\begin{aligned} C(\theta_{n+1}) &\overset{(a)}{\leq} C(\theta_n) - \eta\langle \widehat{\nabla} C_n, d_n\rangle + \eta\langle \widehat{\nabla} C_n - \nabla C(\theta_n), d_n\rangle + \frac{\varepsilon\ell\eta^2}{2}\|d_n\|^2 \\ &\overset{(b)}{\leq} C(\theta_n) + \varepsilon\eta\zeta\|d_n\| + \frac{\varepsilon\ell\eta^2}{2}\|d_n\|^2, \end{aligned}$$

where (a) adds and subtracts $\widehat{\nabla} C_n$, and (b) uses $\langle \widehat{\nabla} C_n, d_n\rangle \geq 0$ (Lemma 4(ii)), the Cauchy–Schwarz inequality, and Assumption 2. This proves (26).

If $n$ is exceptional, Lemma 3(iv) gives $C(\theta_n) \leq (\gamma + \zeta_C)\varepsilon$, and

$$\begin{aligned} C(\theta_{n+1}) &\overset{(a)}{\leq} C(\theta_n) + \eta\|\nabla C(\theta_n)\|\,\|d_n\| + \frac{\varepsilon\ell\eta^2}{2}\|d_n\|^2 \\ &\overset{(b)}{\leq} (\gamma + \zeta_C)\varepsilon + 2\varepsilon\eta M^2 + 2\varepsilon\ell\eta^2 M^2 \overset{(c)}{\leq} (\gamma + \zeta_C + 3\eta M^2)\varepsilon, \end{aligned}$$

where (a) is (22) with the Cauchy–Schwarz inequality; (b) uses $\|\nabla C(\theta_n)\| \leq \varepsilon M$ and $\|d_n\| \leq 2M$; and (c) uses $\ell\eta \leq \frac{1}{2}$. □

### B.4 Number of RESTORE iterations

**Lemma 6.** *The number of RESTORE iterations satisfies*

$$\frac{\eta\mu^2}{4}\,|\mathcal{I}_{\mathrm{R}}| \leq \frac{C(\theta_0)}{\varepsilon} + \sum_{n \in \mathcal{I}_{\mathrm{O}}} \eta\Big(\zeta\|d_n\| + \frac{\ell\eta}{2}\|d_n\|^2\Big). \tag{27}$$

*Proof.* For $0 \leq n \leq N$, define the potential

$$\Phi_n := \max\Big\{\frac{C(\theta_n)}{\varepsilon} - \gamma - \zeta_C - 3\eta M^2,\ 0\Big\}.$$

We show that, for every $0 \le n < N$,

$$\Phi_{n+1} \le \begin{cases} \Phi_n - \dfrac{\eta\mu^2}{4}, & n \in \mathcal{I}_{\mathrm{R}}, \\ \Phi_n + \eta\Big(\zeta\|d_n\| + \dfrac{\ell\eta}{2}\|d_n\|^2\Big), & n \in \mathcal{I}_{\mathrm{O}} \text{ not exceptional}, \\ 0, & n \in \mathcal{I}_{\mathrm{O}} \text{ exceptional}. \end{cases} \tag{28}$$

Throughout, we use that $x \mapsto \max\{x, 0\}$ is nondecreasing and 1-Lipschitz.

*Case $n \in \mathcal{I}_{\mathrm{R}}$.* By Lemma 3(i) and the third condition in (7),

$$\frac{C(\theta_n)}{\varepsilon} - \gamma - \zeta_C - 3\eta M^2 > 1 - \gamma - 2\zeta_C - 3\eta M^2 \ge 3\eta M^2 \ge \frac{\eta\mu^2}{4},$$

so $\Phi_n = C(\theta_n)/\varepsilon - \gamma - \zeta_C - 3\eta M^2 \ge \eta\mu^2/4$. Combining this with (24),

$$\Phi_{n+1} \le \max\Big\{\Phi_n - \frac{\eta\mu^2}{4},\ 0\Big\} = \Phi_n - \frac{\eta\mu^2}{4}.$$

*Case $n \in \mathcal{I}_{\mathrm{O}}$ not exceptional.* By (26), $C(\theta_{n+1})/\varepsilon$ exceeds $C(\theta_n)/\varepsilon$ by at most $\eta(\zeta\|d_n\| + \frac{\ell\eta}{2}\|d_n\|^2)$, and the claim follows from the monotonicity and 1-Lipschitz continuity of $\max\{\cdot, 0\}$.

*Case $n \in \mathcal{I}_{\mathrm{O}}$ exceptional.* By Lemma 5(ii), $C(\theta_{n+1})/\varepsilon \le \gamma + \zeta_C + 3\eta M^2$, hence $\Phi_{n+1} = 0$.

Summing (28) over $n = 0, \dots, N-1$ and using $\Phi_{n+1} \le \Phi_n$ in the exceptional case,

$$0 \le \Phi_N \le \Phi_0 - \frac{\eta\mu^2}{4}|\mathcal{I}_{\mathrm{R}}| + \sum_{n \in \mathcal{I}_{\mathrm{O}}} \eta\Big(\zeta\|d_n\| + \frac{\ell\eta}{2}\|d_n\|^2\Big).$$

Since $C \ge 0$ and $\gamma + \zeta_C + 3\eta M^2 \ge 0$, we have $\Phi_0 \le C(\theta_0)/\varepsilon$, which yields (27). □

### B.5 Proof of Theorem 1

**Part (a).** The first claim is Lemma 3(ii). Suppose $N > 4C(\theta_0)/(\varepsilon\eta\mu^2)$. Then $|\mathcal{I}_{\mathrm{O}}| > 0$ by part (b), whose proof below does not use part (a). If $n_0 = 0$, there is nothing to prove. Otherwise $0, \dots, n_0 - 1 \in \mathcal{I}_{\mathrm{R}}$, and applying (24) $n_0 - 1$ times and Lemma 3(i) at $n = n_0 - 1$,

$$(1 - \zeta_C)\varepsilon < C(\theta_{n_0-1}) \le C(\theta_0) - (n_0 - 1)\frac{\varepsilon\eta\mu^2}{4}.$$

Since $\zeta_C < 1$ by (7), this gives $n_0 < 1 + 4C(\theta_0)/(\varepsilon\eta\mu^2)$.

We prove $C(\theta_n) \le (1 + \zeta_C + 2\eta M(\zeta + \ell\eta M))\varepsilon$ for $n_0 \le n \le N$ by induction. The base case $n = n_0$ follows from Lemma 3(ii). Assume the claim for some $n_0 \le n < N$. If $n \in \mathcal{I}_{\mathrm{R}}$, then $C(\theta_{n+1}) < C(\theta_n)$ by (24). If $n \in \mathcal{I}_{\mathrm{O}}$ is not exceptional, then Lemma 3(ii), (26), and $\|d_n\| \le 2M$ give

$$C(\theta_{n+1}) \le (1 + \zeta_C)\varepsilon + \varepsilon\eta\big(2M\zeta + 2\ell\eta M^2\big) = \big(1 + \zeta_C + 2\eta M(\zeta + \ell\eta M)\big)\varepsilon.$$

If $n \in \mathcal{I}_{\mathrm{O}}$ is exceptional, Lemma 5(ii) and the third condition in (7) give $C(\theta_{n+1}) \le (\gamma + \zeta_C + 3\eta M^2)\varepsilon \le (1 - \zeta_C)\varepsilon$. In all cases the claim holds at $n + 1$.

**Part (b).** We first bound $|\mathcal{I}_{\mathrm{O}}|$. By Lemma 6 and $\|d_n\| \le 2M$,

$$\frac{\eta\mu^2}{4}|\mathcal{I}_{\mathrm{R}}| \le \frac{C(\theta_0)}{\varepsilon} + |\mathcal{I}_{\mathrm{O}}|\,\eta\big(2M\zeta + 2\ell\eta M^2\big),$$

and hence

$$|\mathcal{I}_{\mathrm{R}}| \le \frac{4C(\theta_0)}{\varepsilon\eta\mu^2} + \frac{8M\zeta + 8\ell\eta M^2}{\mu^2}\,|\mathcal{I}_{\mathrm{O}}| \le \frac{4C(\theta_0)}{\varepsilon\eta\mu^2} + \frac{2}{3}|\mathcal{I}_{\mathrm{O}}|,$$

where the last step uses $8M\zeta \le \mu^2/2$ and $8\ell\eta M^2 \le \mu^2/6$ from (7). Since $N = |\mathcal{I}_{\mathrm{O}}| + |\mathcal{I}_{\mathrm{R}}|$, we obtain $|\mathcal{I}_{\mathrm{O}}| \ge \frac{3}{5}\big(N - 4C(\theta_0)/(\varepsilon\eta\mu^2)\big)$.

Next, summing (25) over $n \in \mathcal{I}_{\mathrm{O}}$ and the second inequality in (24) over $n \in \mathcal{I}_{\mathrm{R}}$, and using $L(\theta_N) \ge 0$,

$$\begin{aligned}
\frac{\eta}{2}\sum_{n\in\mathcal{I}_{\mathrm{O}}}\|d_n\|^2 &\overset{(a)}{\le} L(\theta_0) + \eta\zeta^2|\mathcal{I}_{\mathrm{O}}| + 3\eta M^2|\mathcal{I}_{\mathrm{R}}| \\
&\overset{(b)}{\le} L(\theta_0) + \eta\zeta^2|\mathcal{I}_{\mathrm{O}}| + \frac{12M^2}{\mu^2}\cdot\frac{C(\theta_0)}{\varepsilon} + \sum_{n\in\mathcal{I}_{\mathrm{O}}}\Big(\frac{12M^2}{\mu^2}\,\eta\zeta\|d_n\| + \frac{6M^2\ell\eta}{\mu^2}\,\eta\|d_n\|^2\Big) \\
&\overset{(c)}{\le} L(\theta_0) + \frac{12M^2}{\mu^2}\cdot\frac{C(\theta_0)}{\varepsilon} + \eta\zeta^2|\mathcal{I}_{\mathrm{O}}| + \sum_{n\in\mathcal{I}_{\mathrm{O}}}\Big(\frac{\eta}{8}\|d_n\|^2 + \frac{288M^4}{\mu^4}\eta\zeta^2 + \frac{\eta}{8}\|d_n\|^2\Big),
\end{aligned}$$

where (a) telescopes the one-step inequalities; (b) multiplies (27) by $12M^2/\mu^2$; and (c) uses $ab \le \frac{a^2}{8} + 2b^2$ with $a = \|d_n\|$ and $b = 12M^2\zeta/\mu^2$ for the first summand, and $6M^2\ell\eta/\mu^2 \le \frac{1}{8}$ from (7) for the second. Moving the terms $\frac{\eta}{4}\sum_{n\in\mathcal{I}_{\mathrm{O}}}\|d_n\|^2$ to the left-hand side,

$$\frac{\eta}{4}\sum_{n\in\mathcal{I}_{\mathrm{O}}}\|d_n\|^2 \le L(\theta_0) + \frac{12M^2}{\mu^2}\cdot\frac{C(\theta_0)}{\varepsilon} + \eta\zeta^2\Big(1 + \frac{288M^4}{\mu^4}\Big)|\mathcal{I}_{\mathrm{O}}|,$$

and dividing by $\eta|\mathcal{I}_{\mathrm{O}}|/4$ yields (8).

**Part (c).** Let $n \in \mathcal{I}_{\mathrm{O}}$ and let $\nu_n$ be as in Lemma 4(iii). Since $d_n = \widehat{\nabla}L_n + \nu_n\widehat{\nabla}C_n$,

$$\nabla L(\theta_n) + \nu_n\nabla C(\theta_n) = d_n + \big(\nabla L(\theta_n) - \widehat{\nabla}L_n\big) + \nu_n\big(\nabla C(\theta_n) - \widehat{\nabla}C_n\big),$$

and Assumption 2 gives

$$\|\nabla L(\theta_n) + \nu_n\nabla C(\theta_n)\| \le \|d_n\| + \zeta + \varepsilon\nu_n\zeta. \tag{29}$$

If $n$ is not projected, $\nu_n = 0$ and (9) follows from (29). If $n$ is projected, Lemma 3(iii) and Assumption 3 give $\|\nabla C(\theta_n)\| \ge \varepsilon\mu$, so $\|\widehat{\nabla}C_n\| \ge \varepsilon(\mu - \zeta) \ge \varepsilon\mu/2$. Hence

$$\varepsilon\nu_n = \frac{\varepsilon\,|\langle\widehat{\nabla}C_n, \widehat{\nabla}L_n\rangle|}{\|\widehat{\nabla}C_n\|^2} \le \frac{\varepsilon\|\widehat{\nabla}L_n\|}{\|\widehat{\nabla}C_n\|} \le \frac{2M}{\mu/2} = \frac{4M}{\mu},$$

using (23). Substituting into (29) and using $\mu \le M$ gives the first inequality in (9), since $\zeta(1 + 4M/\mu) \le 5M\zeta/\mu$. For complementary slackness, if $\nu_n = 0$ there is nothing to prove. If $\nu_n > 0$, the iteration is projected, so $(\gamma - \zeta_C)\varepsilon < C(\theta_n) \le (1 + \zeta_C)\varepsilon$ by Lemma 3(ii)–(iii), and

$$-\varepsilon\nu_n\zeta_C \le \nu_n\big(\varepsilon - C(\theta_n)\big) < \varepsilon\nu_n\,(1 - \gamma + \zeta_C).$$

Since $\zeta_C \le 1 - \gamma + \zeta_C$ and $\varepsilon\nu_n \le 4M/\mu$, the last inequality in (9) follows. Finally, $\|d_{n^*}\|^2 = \min_{n\in\mathcal{I}_{\mathrm{O}}}\|d_n\|^2$ is at most the average in (8), so the first inequality in (9) with $n = n^*$ gives the stationarity residual $r_{\mathrm{S}}$ in (10). Part (a) gives $C(\theta_{n^*}) \le (1 + \zeta_C)\varepsilon$, and the last inequality in (9) gives the complementarity residual $r_{\mathrm{C}}$. Hence $\theta_{n^*}$ is an $(r_{\mathrm{S}}, \zeta_C, r_{\mathrm{C}})$-KKT point with multiplier $\nu_{n^*} \ge 0$. □

### B.6 Tightness of the complementarity residual

The complementarity residual $r_{\mathrm{C}}$ in (10) does not decrease with $N$. The following example shows that its dependence on $1-\gamma$ cannot be removed, even with exact gradients and exact constraint values.

**Proposition 1.** *Let $D = 1$, $L(\theta) = \frac{1}{2}(\theta-2)^2$ and $C(\theta) = \frac{\varepsilon}{2}\theta^2$. Run Algorithm 1 with exact quantities $\widehat{\nabla} L_n = \nabla L(\theta_n)$, $\widehat{\nabla} C_n = \nabla C(\theta_n)$ and $\widehat{C}_n = C(\theta_n)$, initial point $\theta_0 = 0$, gate $\gamma \in (0,1)$, and step size $0 < \eta \le \min\{\gamma/96, (1-\gamma)/24\}$. Then the following hold.*

*(i) The hypotheses of Theorem 1 hold with $\Theta = [0, \sqrt{2}]$, $\ell = 1$, $M = 2$, $\mu = \sqrt{2\gamma}$ and $\zeta = \zeta_C = 0$.*

*(ii) There exist $\bar{n} \ge 1$ and $\bar{\theta} \in (\sqrt{2\gamma}, \sqrt{2})$ such that $\theta_n = \bar{\theta}$, $n \in \mathcal{I}_{\mathrm{O}}$ and $d_n = 0$ for all $n \ge \bar{n}$, while $d_n \neq 0$ for all $n < \bar{n}$.*

*(iii) The unique $\nu \in \mathbb{R}$ with $\nabla L(\bar{\theta}) + \nu \nabla C(\bar{\theta}) = 0$ is $\nu = (2-\bar{\theta})/(\varepsilon\bar{\theta}) > 0$, and*

$$\nu\big(\varepsilon - C(\bar{\theta})\big) \ge \frac{1-\gamma}{7}. \tag{30}$$

*Consequently, if $\bar{\theta}$ is an $(r_{\mathrm{S}}, r_{\mathrm{F}}, r_{\mathrm{C}})$-KKT point of (5), then $r_{\mathrm{C}} \ge (1-\gamma)/7 - r_{\mathrm{S}}/\sqrt{2\gamma}$.*

*In particular, for every $N > \bar{n}$, the output is $\theta_{n^*} = \bar{\theta}$; it satisfies stationarity and feasibility exactly, but its complementarity residual is at least $(1-\gamma)/7$.*

*Proof.* We first prove (ii). Suppose $\theta_n \in [0, \sqrt{2\gamma}]$. Then $C(\theta_n) \le \gamma\varepsilon$, so $n \in \mathcal{I}_{\mathrm{O}}$ and the projection is not applied. Hence $d_n = \nabla L(\theta_n) = \theta_n - 2 \neq 0$ and

$$\theta_{n+1} = \theta_n + \eta(2-\theta_n) \in \big[\theta_n,\ \theta_n + 2\eta\big].$$

As long as the iterates remain in $[0, \sqrt{2\gamma}]$, this recursion gives $\theta_n = 2 - 2(1-\eta)^n$, which exceeds $\sqrt{2\gamma} < 2$ for large $n$. Hence there is a first index $\bar{n} \ge 1$ with $\theta_{\bar{n}} > \sqrt{2\gamma}$, and $d_n \neq 0$ for $n < \bar{n}$. Let $\bar{\theta} := \theta_{\bar{n}}$. Since $\theta_{\bar{n}-1} \le \sqrt{2\gamma}$,

$$\sqrt{2\gamma} < \bar{\theta} \le \sqrt{2\gamma} + 2\eta \le \sqrt{2\gamma} + \frac{1-\gamma}{12} \le \frac{\sqrt{2} + \sqrt{2\gamma}}{2} < \sqrt{2}, \tag{31}$$

where the third inequality uses $\eta \le (1-\gamma)/24$, and the fourth uses $\frac{1}{2}(\sqrt{2} - \sqrt{2\gamma}) = \frac{1-\gamma}{\sqrt{2}(1+\sqrt{\gamma})} \ge \frac{1-\gamma}{2\sqrt{2}} \ge \frac{1-\gamma}{12}$. By (31), $\gamma\varepsilon < C(\bar{\theta}) < \varepsilon$, so $\bar{n} \in \mathcal{I}_{\mathrm{O}}$ and $\widehat{C}_{\bar{n}} > \gamma\varepsilon$. Moreover, $\langle \nabla C(\bar{\theta}), \nabla L(\bar{\theta}) \rangle = \varepsilon\bar{\theta}(\bar{\theta} - 2) < 0$, so the projection is applied. Since $D = 1$ and $\nabla C(\bar{\theta}) = \varepsilon\bar{\theta} \neq 0$, the component of $\nabla L(\bar{\theta})$ orthogonal to $\nabla C(\bar{\theta})$ is zero, i.e., $d_{\bar{n}} = 0$ and $\theta_{\bar{n}+1} = \bar{\theta}$. By induction, $\theta_n = \bar{\theta}$, $n \in \mathcal{I}_{\mathrm{O}}$ and $d_n = 0$ for all $n \ge \bar{n}$.

For (i), all iterates lie in $[0, \bar{\theta}] \subset \Theta$. On $\Theta$, $\nabla L(\theta) = \theta - 2$ is 1-Lipschitz with $|\nabla L(\theta)| \le 2$, and $\nabla C(\theta) = \varepsilon\theta$ is $\varepsilon$-Lipschitz with $|\nabla C(\theta)| \le \sqrt{2}\,\varepsilon \le 2\varepsilon$, so Assumption 1 holds with $\ell = 1$ and $M = 2$. Assumption 2 holds with $\zeta = \zeta_C = 0$. If $\theta \in \Theta$ and $C(\theta) \ge \gamma\varepsilon$, then $\theta \ge \sqrt{2\gamma}$ and $|\nabla C(\theta)| \ge \varepsilon\sqrt{2\gamma}$, so Assumption 3 holds with $\mu = \sqrt{2\gamma}$. Finally, (7) reads $\zeta = 0$, $\eta \le \gamma/96 = \mu^2/(48M^2)$, $24\eta \le 1-\gamma$ and $0 < \gamma$, all of which hold.

For (iii), the equation $\bar{\theta} - 2 + \nu\varepsilon\bar{\theta} = 0$ has the unique solution $\nu = (2-\bar{\theta})/(\varepsilon\bar{\theta})$, which is positive because $\bar{\theta} \in (0, 2)$. Then

$$\begin{aligned}\nu\big(\varepsilon - C(\bar{\theta})\big) = \frac{2-\bar{\theta}}{\bar{\theta}}\Big(1 - \frac{\bar{\theta}^2}{2}\Big) &\overset{(a)}{\ge} (\sqrt{2}-1)\Big(1 - \frac{(1+\sqrt{\gamma})^2}{4}\Big) \overset{(b)}{=} (\sqrt{2}-1)\,\frac{(1-\sqrt{\gamma})(3+\sqrt{\gamma})}{4} \\ &\overset{(c)}{\ge} \frac{3(\sqrt{2}-1)}{8}\,(1-\gamma) \ge \frac{1-\gamma}{7},\end{aligned}$$

where (a) uses $\bar\theta < \sqrt{2}$, which gives $(2-\bar\theta)/\bar\theta > \sqrt{2}-1$, and $\bar\theta \le (\sqrt{2}+\sqrt{2\gamma})/2 = (1+\sqrt{\gamma})/\sqrt{2}$ from (31), which gives $\bar\theta^2/2 \le (1+\sqrt{\gamma})^2/4$; (b) uses $4-(1+\sqrt{\gamma})^2 = (1-\sqrt{\gamma})(3+\sqrt{\gamma})$; (c) uses $3+\sqrt{\gamma} \ge 3$ and $1-\sqrt{\gamma} = (1-\gamma)/(1+\sqrt{\gamma}) \ge (1-\gamma)/2$; and the last step uses $3(\sqrt{2}-1)/8 > 1/7$. This proves (30). For the last claim of (iii), let $\nu' \ge 0$ satisfy (6) at $\bar\theta$. Since $|\nabla L(\bar\theta)+\nu'\nabla C(\bar\theta)| = \varepsilon\bar\theta\,|\nu'-\nu| \le r_{\mathrm{S}}$ and $0 < \varepsilon - C(\bar\theta) \le \varepsilon$,

$$r_{\mathrm{C}} \ge \nu'\big(\varepsilon - C(\bar\theta)\big) \ge \nu\big(\varepsilon - C(\bar\theta)\big) - |\nu'-\nu|\,\varepsilon \ge \frac{1-\gamma}{7} - \frac{r_{\mathrm{S}}}{\bar\theta} \ge \frac{1-\gamma}{7} - \frac{r_{\mathrm{S}}}{\sqrt{2\gamma}}.$$

The final statement follows from (ii) and (iii): for $N > \bar n$, exactly the indices $n \ge \bar n$ attain $\min_{n\in\mathcal{I}_{\mathrm{O}}} \|d_n\| = 0$, so $\theta_{n^*} = \bar\theta$, and $C(\bar\theta) < \varepsilon$. □

# C Proof of Corollary 1 and Estimating the Constraint

**Proof of Corollary 1.** We first show that, with probability at least $1-q$, Assumption 2 holds with

$$\zeta = \frac{r\mu^3}{100M^2}, \qquad \zeta_C = \frac{r\mu}{32M}. \tag{32}$$

Fix $0 \le n < N$ and condition on $\mathcal{F}_n$, so that $\theta_n$ is fixed. The error $\widehat\nabla L_n - \nabla L(\theta_n)$ is the average of $|B_n|$ independent, zero-mean, norm-subGaussian vectors with parameter $\sigma$. By the Hoeffding-type inequality for norm-subGaussian vectors [6, Corollary 7], there is an absolute constant $c_J$ such that, for any $q_0 \in (0,1)$, with conditional probability at least $1-q_0$,

$$\|\widehat\nabla L_n - \nabla L(\theta_n)\| \le c_J\,\sigma\sqrt{\frac{\log(2D/q_0)}{|B_n|}}.$$

The same bound holds for the relative gradient error $\|\widehat\nabla C_n - \nabla C(\theta_n)\|/\varepsilon$ of the constraint. It also holds for the relative estimation error $|\widehat C_n - C(\theta_n)|/\varepsilon$ with $\sigma_C$ in place of $\sigma$, since $\log(2/q_0) \le \log(2D/q_0)$. We take $q_0 = q/(3N)$ and $c_0 = c_J^2$, so that $\log(2D/q_0) = \log(6DN/q)$. The condition on $|B_n|$ in (11) then makes the three right-hand sides at most $r\mu^3/(100M^2)$, $r\mu^3/(100M^2)$ and $r\mu/(32M)$, respectively. Taking expectations of the conditional probabilities and applying a union bound over the $3N$ events proves the claim.

On this event, we verify (7). Using $r \le 1$ and $\mu \le M$,

$$\zeta = \frac{r\mu^3}{100M^2} \le \frac{\mu^2}{100M} \le \frac{\mu^2}{16M}, \qquad \ell\eta \le \frac{\mu^2}{48M^2},$$
$$2\zeta_C + 6\eta M^2 \le \frac{r\mu}{16M} + 6M^2\cdot\frac{r\mu}{96M^3} = \frac{r\mu}{8M} = 1-\gamma, \qquad \zeta_C \le \frac{1}{32} < \frac{7}{8} \le \gamma.$$

Hence Theorem 1 applies. By the first condition in (12) and Theorem 1(b), $|\mathcal{I}_{\mathrm{O}}| \ge \frac{3}{5}\cdot\frac{N}{2} = \frac{3N}{10}$; in particular, $\mathcal{I}_{\mathrm{O}} \ne \emptyset$. We now bound the three residuals in (10).

*Stationarity.* The term under the square root in $r_{\mathrm{S}}$ satisfies

$$\frac{4}{\eta|\mathcal{I}_{\mathrm{O}}|}\Big(L(\theta_0) + \frac{12M^2}{\mu^2}\cdot\frac{C(\theta_0)}{\varepsilon}\Big) + 4\zeta^2\Big(1+\frac{288M^4}{\mu^4}\Big) \overset{(a)}{\le} \frac{40}{3\eta N}\Big(L(\theta_0) + \frac{12M^2}{\mu^2}\cdot\frac{C(\theta_0)}{\varepsilon}\Big) + \frac{1156\,M^4\zeta^2}{\mu^4}$$
$$\overset{(b)}{\le} \frac{r^2\mu^2}{8} + \frac{1156\,r^2\mu^2}{10^4} \le \frac{r^2\mu^2}{4},$$

where (a) uses $|\mathcal{I}_{\mathrm{O}}| \ge 3N/10$ and $1 + 288M^4/\mu^4 \le 289M^4/\mu^4$, and (b) uses the second condition in (12) and (32). Moreover, $5M\zeta/\mu = r\mu^2/(20M) \le r\mu/20$. Hence $r_{\mathrm{S}} \le r\mu/2 + r\mu/20 \le r\mu$.

*Primal feasibility.* The feasibility residual is $\zeta_C = r\mu/(32M) \le r/32$.
*Complementary slackness.* By (11) and (32),

$$r_{\mathrm{C}} = \frac{4M}{\mu}\Big(\frac{r\mu}{8M} + \frac{r\mu}{32M}\Big) = \frac{5r}{8} \le r.$$

By Theorem 1, $\theta_{n^*}$ is therefore an $(r\mu, r/32, r)$-KKT point of (5). □

*Remark* 13 (Exponential moving average). The practical algorithm uses the exponential moving average $\widehat{C}_n = \beta\widehat{C}_{n-1} + (1-\beta)C_{B_n}(\theta_n)$, where $C_{B_n}$ is the minibatch mean of $c$ and $\widehat{C}_{-1}$ is an initial estimate. Since the weights satisfy $\sum_{j=0}^{n}(1-\beta)\beta^{n-j} = 1-\beta^{n+1}$, the estimation error decomposes as

$$\widehat{C}_n - C(\theta_n) = \sum_{j=0}^{n}(1-\beta)\beta^{n-j}\big(C_{B_j}(\theta_j) - C(\theta_j)\big) + \sum_{j=0}^{n}(1-\beta)\beta^{n-j}\big(C(\theta_j) - C(\theta_n)\big) + \beta^{n+1}\big(\widehat{C}_{-1} - C(\theta_n)\big).$$

Under the conditional value-noise assumptions of Corollary 1, the first term is a weighted sum of martingale differences whose squared weights sum to at most $(1-\beta)/(1+\beta)$; it concentrates at the scale of the minibatch noise of $c$ multiplied by $\sqrt{(1-\beta)/(1+\beta)}$, and it does not decrease with $\eta$. The second term is a lag: since $\|\theta_{i+1} - \theta_i\| = \eta\|d_i\| \le 2\eta M$ (Lemma 5) and $\|\nabla C\| \le \varepsilon M$ on $\Theta$, we have $|C(\theta_j) - C(\theta_n)| \le 2\varepsilon\eta M^2(n-j)$, so the lag is at most $2\varepsilon\eta M^2 \sum_{i\ge 0}(1-\beta)\beta^i i = 2\varepsilon\eta M^2\beta/(1-\beta)$. The third term decays geometrically. A bound on the noise term that holds uniformly over $0 \le n < N$ costs an additional $\sqrt{\log N}$ factor. Hence $\zeta_C$ is at most the sum of a noise term, $2\eta M^2\beta/(1-\beta)$, and a transient. Because the third condition in (7) requires $2\zeta_C < 1-\gamma$, the noise of $c$ must be small compared with the margin $(1-\gamma)\varepsilon$. A larger $\beta$ reduces the noise term but increases the lag, which must then be compensated by a smaller $\eta$; alternatively, $\widehat{C}_n$ can be computed from a separate, larger batch.

# D Proof of Corollary 2

**Lemma 7** (Pathwise error identity)**.** *Fix $\theta$, a window $w$, and a conditioning gravity $g'$, and suppose that the true positions $p^\star_{k,t-1}, \dots, p^\star_{k,t+H}$ follow free fall under $g$ before $t$ and $g'$ after $t$, as in Lemma 1. Then, for every object $k$ and $s = 0, 1, \dots, H$,*

$$e_{k,t+s} = (1+s)\,e_{k,t} - s\,e_{k,t-1} + \Delta t^2 \sum_{j=0}^{s-1}(s-j)\big(a_{k,t+j} - \bar{g}_{t+j}\big). \tag{33}$$

*Consequently, for $s = 1, \dots, H$,*

$$\frac{1}{K}\sum_{k=1}^{K}\|e_{k,t+s}\| \le sH\Delta t^2\sqrt{c} + \frac{1}{K}\sum_{k=1}^{K}\Big((1+s)\|e_{k,t}\| + s\|e_{k,t-1}\|\Big). \tag{34}$$

*Proof.* By Lemma 1, $p^\star_{k,\tau+1} - 2p^\star_{k,\tau} + p^\star_{k,\tau-1} = \bar{g}_\tau\Delta t^2$ for $t \le \tau \le t+H-1$. Subtracting this from the definition of $a_{k,\tau}$,

$$e_{k,\tau+1} = 2e_{k,\tau} - e_{k,\tau-1} + \Delta t^2\big(a_{k,\tau} - \bar{g}_\tau\big), \qquad t \le \tau \le t+H-1. \tag{35}$$

We prove (33) by induction on $s$. For $s = 0$ it reads $e_{k,t} = e_{k,t}$, and for $s = 1$ it is (35) with $\tau = t$. Suppose (33) holds for $s-1$ and $s$, where $1 \le s \le H-1$. By (35) with $\tau = t+s$,

$$\begin{aligned} e_{k,t+s+1} &= 2e_{k,t+s} - e_{k,t+s-1} + \Delta t^2\big(a_{k,t+s} - \bar g_{t+s}\big) \\ &= (2+s)\, e_{k,t} - (1+s)\, e_{k,t-1} \\ &\quad + \Delta t^2\Big(2\sum_{j=0}^{s-1}(s-j)\big(a_{k,t+j} - \bar g_{t+j}\big) - \sum_{j=0}^{s-2}(s-1-j)\big(a_{k,t+j} - \bar g_{t+j}\big) + \big(a_{k,t+s} - \bar g_{t+s}\big)\Big) \\ &= (2+s)\, e_{k,t} - (1+s)\, e_{k,t-1} + \Delta t^2 \sum_{j=0}^{s}(s+1-j)\big(a_{k,t+j} - \bar g_{t+j}\big), \end{aligned}$$

where the second equality substitutes the induction hypothesis for $s$ and $s-1$, and the third collects coefficients: $2(s-j) - (s-1-j) = s+1-j$ for $0 \le j \le s-2$, the coefficient of the summand $j = s-1$ is $2 = s+1-j$, and that of $j = s$ is $1 = s+1-j$. This is (33) for $s+1$.

For (34), the triangle inequality applied to (33) gives

$$\begin{aligned} \frac{1}{K}\sum_{k=1}^{K}\|e_{k,t+s}\| &\overset{(a)}{\le} \frac{\Delta t^2}{K}\sum_{k=1}^{K}\sum_{j=0}^{s-1}(s-j)\|a_{k,t+j} - \bar g_{t+j}\| + \frac{1}{K}\sum_{k=1}^{K}\Big((1+s)\|e_{k,t}\| + s\|e_{k,t-1}\|\Big) \\ &\overset{(b)}{\le} \frac{s\Delta t^2}{K}\sum_{k=1}^{K}\sum_{j=0}^{H-1}\|a_{k,t+j} - \bar g_{t+j}\| + \frac{1}{K}\sum_{k=1}^{K}\Big((1+s)\|e_{k,t}\| + s\|e_{k,t-1}\|\Big) \\ &\overset{(c)}{\le} \frac{s\Delta t^2}{K}\sqrt{KH\sum_{k=1}^{K}\sum_{j=0}^{H-1}\|a_{k,t+j} - \bar g_{t+j}\|^2} + \frac{1}{K}\sum_{k=1}^{K}\Big((1+s)\|e_{k,t}\| + s\|e_{k,t-1}\|\Big), \end{aligned}$$

where (a) is the triangle inequality; (b) uses $s-j \le s$ and $s \le H$; and (c) is the Cauchy–Schwarz inequality over the $KH$ summands. By (3), the square root equals $KH\sqrt{c}$, which yields (34). □

**Proof of Corollary 2.** Fix a predictor $\theta$ satisfying the constraint in the statement. The joint density bound gives $\mathbb{E}_{\mathbb{Q}}[c] \le \kappa C(\theta)$ and $\mathbb{E}_{\mathbb{Q}}\|e_{k,\tau}\|^2 \le \kappa\delta^2$ for $\tau = t-1, t$, since $c$ and the two anchor errors are functions of the variables in that joint law. Applying the $L^2$ triangle inequality to (33) over the test distribution and the average over objects gives

$$\begin{aligned} \left(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\|e_{k,t+s}\|^2\right)^{1/2} &\le (1+s)\left(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\|e_{k,t}\|^2\right)^{1/2} + s\left(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\|e_{k,t-1}\|^2\right)^{1/2} \\ &\quad + \Delta t^2\left(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\left\|\sum_{j=0}^{s-1}(s-j)(a_{k,t+j} - \bar g_{t+j})\right\|^2\right)^{1/2} \\ &\le (2s+1)\sqrt{\kappa}\,\delta + \Delta t^2\left(\frac{s(s+1)(2s+1)}{6K}\sum_{k=1}^{K}\sum_{j=0}^{H-1}\mathbb{E}_{\mathbb{Q}}\|a_{k,t+j} - \bar g_{t+j}\|^2\right)^{1/2} \\ &\le \sqrt{\kappa}\left[(2s+1)\delta + \Delta t^2\sqrt{\frac{Hs(s+1)(2s+1)}{6}\,C(\theta)}\right]. \end{aligned}$$

The second inequality uses the Cauchy–Schwarz inequality for the sum over time, the identity $\sum_{j=0}^{s-1}(s-j)^2 = s(s+1)(2s+1)/6$, and $s \le H$; the last uses (3) and the density bound. All these $L^2$

quantities are finite by the same bounds. Substituting $C(\theta) \le (1+\zeta_C)\varepsilon$ and $\varepsilon \le 16(1+\rho)\delta^2/\Delta t^4$ from Lemma 2 proves the second inequality in (13). For a random predictor independent of the test data, the argument applies conditionally on its value. No independence between errors at different times is used. □

# E Proof of Theorem 2

Throughout this appendix, the hypotheses in the preamble of Theorem 2 hold and $s \in \{1, \dots, H\}$ is fixed.

**Part (a).** Under $\mathbb{P}$ and under $\mathbb{Q}$, the window has the law of real windows and $g'$ is independent of it. The context $x_{\le t}$, the gravity $g$ and the true positions $p^\star_{k,t-1}, p^\star_{k,t}$ are determined by the window. Hence the joint law of $(x_{\le t}, g, g', (p^\star_{k,t-1}, p^\star_{k,t})_{k=1}^K)$ under $\mathbb{Q}$ has a density with respect to its law under $\mathbb{P}$, namely the density of the law of $g'$ under $\mathbb{Q}$ with respect to $\mathcal{U}(\mathcal{G})$, which is at most $\kappa$. The real windows are contact-free, and the counterfactual trajectories are contact-free by hypothesis. Since $n^* \in \mathcal{I}_{\mathrm{O}}$, Theorem 1(a) gives $C(\theta_{n^*}) \le (1+\zeta_C)\varepsilon$. Therefore Corollary 2 applies to $\theta = \theta_{n^*}$, and its second inequality, with $p_{k,t+s} = p_{k,t+s}(\theta_{n^*}, g')$ and $p^\star_{k,t+s} = p^\star_{k,t+s}(g')$, is (16). □

**Part (b).** We first bound the length of the path traced by continued JEPA training.

**Lemma 8.** *Under the hypotheses of Theorem 2(b), for every $0 \le n \le N$,*

$$\sum_{i=0}^{n-1} \|\theta_{i+1}^{\mathrm{JEPA}} - \theta_i^{\mathrm{JEPA}}\| \le \sqrt{2\eta n\Big(L(\theta_0) - \inf_{\theta\in\mathbb{R}^D} L(\theta)\Big)}. \tag{36}$$

*Proof.* For $n = 0$, both sides vanish. Let $1 \le n \le N$ and $0 \le i < n$. Applying (22) to $L$ on the segment $[\theta_i^{\mathrm{JEPA}}, \theta_{i+1}^{\mathrm{JEPA}}] \subset \Theta'$ with $d = \nabla L(\theta_i^{\mathrm{JEPA}})$,

$$L(\theta_{i+1}^{\mathrm{JEPA}}) \overset{(a)}{\le} L(\theta_i^{\mathrm{JEPA}}) - \eta\|\nabla L(\theta_i^{\mathrm{JEPA}})\|^2 + \frac{\ell\eta^2}{2}\|\nabla L(\theta_i^{\mathrm{JEPA}})\|^2 \overset{(b)}{\le} L(\theta_i^{\mathrm{JEPA}}) - \frac{\eta}{2}\|\nabla L(\theta_i^{\mathrm{JEPA}})\|^2,$$

where (a) is (22) with the Lipschitz constant $\ell$ of $\nabla L$ on $\Theta'$, and (b) uses $\eta\ell \le 1$. Summing over $i = 0, \dots, n-1$ and using $L(\theta_n^{\mathrm{JEPA}}) \ge \inf_{\theta\in\mathbb{R}^D} L(\theta)$, which is finite because $L \ge 0$,

$$\frac{\eta}{2}\sum_{i=0}^{n-1} \|\nabla L(\theta_i^{\mathrm{JEPA}})\|^2 \le L(\theta_0) - L(\theta_n^{\mathrm{JEPA}}) \le L(\theta_0) - \inf_{\theta\in\mathbb{R}^D} L(\theta).$$

Hence, by (14),

$$\begin{aligned}\sum_{i=0}^{n-1} \|\theta_{i+1}^{\mathrm{JEPA}} - \theta_i^{\mathrm{JEPA}}\| = \eta\sum_{i=0}^{n-1} \|\nabla L(\theta_i^{\mathrm{JEPA}})\| &\overset{(a)}{\le} \eta\sqrt{n}\Big(\sum_{i=0}^{n-1} \|\nabla L(\theta_i^{\mathrm{JEPA}})\|^2\Big)^{1/2} \\ &\overset{(b)}{\le} \sqrt{2\eta n\Big(L(\theta_0) - \inf_{\theta\in\mathbb{R}^D} L(\theta)\Big)},\end{aligned}$$

where (a) is the Cauchy–Schwarz inequality and (b) uses the previous display. □

We now prove (18). Each root-mean-square expression below is $K^{-1/2}$ times the norm of a $K$-tuple of random vectors in $L^2(\mathbb{Q};\mathbb{R}^{2K})$, and therefore satisfies the Minkowski inequality. Let $0 \le n \le N$. Then

$$
\begin{aligned}
&\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\big\|p_{k,t+s}(\theta_n^{\mathrm{JEPA}},g') - p^{\star}_{k,t+s}(g')\big\|^2\Big)^{1/2}\\
&\quad\overset{(a)}{\ge}\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\big\|p_{k,t+s}(\theta_0,g') - p^{\star}_{k,t+s}(g')\big\|^2\Big)^{1/2}\\
&\qquad-\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\big\|p_{k,t+s}(\theta_n^{\mathrm{JEPA}},g') - p_{k,t+s}(\theta_0,g')\big\|^2\Big)^{1/2}\\
&\quad\overset{(b)}{\ge}\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\big\|p_{k,t+s}(\theta_0,g') - p^{\star}_{k,t+s}(g')\big\|^2\Big)^{1/2}\\
&\qquad-\sum_{i=0}^{n-1}\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\big\|p_{k,t+s}(\theta_{i+1}^{\mathrm{JEPA}},g') - p_{k,t+s}(\theta_i^{\mathrm{JEPA}},g')\big\|^2\Big)^{1/2}\\
&\quad\overset{(c)}{\ge}\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\big\|p_{k,t+s}(\theta_0,g') - p^{\star}_{k,t+s}(g')\big\|^2\Big)^{1/2} - \Lambda\sum_{i=0}^{n-1}\big\|\theta_{i+1}^{\mathrm{JEPA}} - \theta_i^{\mathrm{JEPA}}\big\|\\
&\quad\overset{(d)}{\ge}\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\big\|p_{k,t+s}(\theta_0,g') - p^{\star}_{k,t+s}(g')\big\|^2\Big)^{1/2} - \Lambda\sqrt{2\eta n\Big(L(\theta_0) - \inf_{\theta\in\mathbb{R}^D}L(\theta)\Big)},
\end{aligned}
$$

where (a) is the Minkowski inequality applied to the decomposition

$$
\begin{aligned}
&p_{k,t+s}(\theta_0,g') - p^{\star}_{k,t+s}(g')\\
&\qquad= \big(p_{k,t+s}(\theta_n^{\mathrm{JEPA}},g') - p^{\star}_{k,t+s}(g')\big) - \big(p_{k,t+s}(\theta_n^{\mathrm{JEPA}},g') - p_{k,t+s}(\theta_0,g')\big);
\end{aligned}
$$

(b) is the Minkowski inequality applied to the telescoping sum $p_{k,t+s}(\theta_n^{\mathrm{JEPA}},g') - p_{k,t+s}(\theta_0,g') = \sum_{i=0}^{n-1}\big(p_{k,t+s}(\theta_{i+1}^{\mathrm{JEPA}},g') - p_{k,t+s}(\theta_i^{\mathrm{JEPA}},g')\big)$; (c) is (17), which applies because $\theta_i^{\mathrm{JEPA}},\theta_{i+1}^{\mathrm{JEPA}} \in \Theta'$; and (d) is Lemma 8. The subtracted terms are finite by (c) and (d), so the chain is valid also when the first term is infinite. This proves (18).

Suppose now that $\theta_0$ is gravity-blind. By Lemma 1, the counterfactual trajectory and the actual future of the window share the position $p^{\star}_{k,t}$ and the velocity of object $k$ at step $t$, and follow free fall under $g'$ and under $g$, respectively, after step $t$. Hence

$$
p^{\star}_{k,t+s}(g') - p^{\star}_{k,t+s}(g) = \frac{(s\Delta t)^2}{2}\,(g'-g) \qquad (k=1,\dots,K). \tag{37}
$$

Therefore

$$\begin{aligned}
&\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\big\|p_{k,t+s}(\theta_0,g')-p^{\star}_{k,t+s}(g')\big\|^2\Big)^{1/2}\\
&\quad\overset{(a)}{=}\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\Big\|p_{k,t+s}(\theta_0,g)-p^{\star}_{k,t+s}(g)-\frac{(s\Delta t)^2}{2}(g'-g)\Big\|^2\Big)^{1/2}\\
&\quad\overset{(b)}{\geq}\frac{(s\Delta t)^2}{2}\big(\mathbb{E}_{\mathbb{Q}}\|g'-g\|^2\big)^{1/2}-\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}_{\mathbb{Q}}\big\|p_{k,t+s}(\theta_0,g)-p^{\star}_{k,t+s}(g)\big\|^2\Big)^{1/2}\\
&\quad\overset{(c)}{=}\frac{(s\Delta t)^2}{2}\big(\mathbb{E}_{\mathbb{Q}}\|g'-g\|^2\big)^{1/2}-\Big(\frac{1}{K}\sum_{k=1}^{K}\mathbb{E}\big\|p_{k,t+s}(\theta_0,g)-p^{\star}_{k,t+s}(g)\big\|^2\Big)^{1/2},
\end{aligned}$$

where (a) uses $p_{k,t+s}(\theta_0,g')=p_{k,t+s}(\theta_0,g)$, which holds because $\theta_0$ is gravity-blind, and (37); (b) is the Minkowski inequality, together with the fact that the vector $\frac{1}{2}(s\Delta t)^2(g'-g)$ does not depend on $k$; and (c) holds because $p_{k,t+s}(\theta_0,g)-p^{\star}_{k,t+s}(g)$ is a function of the window, whose law under $\mathbb{Q}$ is that of real windows. This proves (19). □

**Part (c).** We prove $\theta_n^{\text{fact}}=\theta_n^{\text{JEPA}}$ for $0\leq n\leq N$ by induction. For $n=0$, both equal $\theta_0$. Suppose $\theta_i^{\text{fact}}=\theta_i^{\text{JEPA}}$ for $0\leq i\leq n$, where $n<N$. Then

$$\begin{aligned}
C_{\text{fact}}(\theta_n^{\text{fact}})\overset{(a)}{=}C_{\text{fact}}(\theta_n^{\text{JEPA}})&\overset{(b)}{\leq}C_{\text{fact}}(\theta_0)+\varepsilon M\sum_{i=0}^{n-1}\big\|\theta_{i+1}^{\text{JEPA}}-\theta_i^{\text{JEPA}}\big\|\\
&\overset{(c)}{\leq}C_{\text{fact}}(\theta_0)+\varepsilon M\sqrt{2\eta N\Big(L(\theta_0)-\inf_{\theta\in\mathbb{R}^D}L(\theta)\Big)}\overset{(d)}{\leq}\gamma\varepsilon,
\end{aligned}$$

where (a) is the induction hypothesis; (b) applies the mean value theorem on each segment $[\theta_i^{\text{JEPA}},\theta_{i+1}^{\text{JEPA}}]\subset\Theta'$, on which $\|\nabla C_{\text{fact}}\|\leq\varepsilon M$; (c) is Lemma 8 with $n\leq N$; and (d) is (20). Since the estimate is exact, $\widehat{C}_n=C_{\text{fact}}(\theta_n^{\text{fact}})\leq\gamma\varepsilon<\varepsilon$. Hence iteration $n$ is an OPTIMISE iteration at which the gate in line 7 fails, so $d_n=\nabla L(\theta_n^{\text{fact}})=\nabla L(\theta_n^{\text{JEPA}})$ and

$$\theta_{n+1}^{\text{fact}}=\theta_n^{\text{JEPA}}-\eta\nabla L(\theta_n^{\text{JEPA}})=\theta_{n+1}^{\text{JEPA}}$$

by (14). This completes the induction. Every iteration of the factual-constraint method is an OPTIMISE iteration, so its output is $\theta_n^{\text{JEPA}}$ for some $0\leq n<N$, and the bounds in (b) apply to it. □

**Verification of the example in Remark 10.** Let $D=1$, $L(\theta)=\frac{1}{2}(\theta-a)^2$ with $a>\sqrt{2}$, and $C(\theta)=\frac{\varepsilon}{2}\theta^2$. For $\lambda\geq 0$, the function $L+\lambda C$ is strongly convex, and its unique stationary point solves $\theta-a+\lambda\varepsilon\theta=0$, i.e., $\theta=a/(1+\lambda\varepsilon)$. At this point,

$$C\Big(\frac{a}{1+\lambda\varepsilon}\Big)=\frac{\varepsilon a^2}{2(1+\lambda\varepsilon)^2}\leq\varepsilon\quad\Longleftrightarrow\quad a\leq\sqrt{2}\,(1+\lambda\varepsilon)\quad\Longleftrightarrow\quad\lambda\geq\frac{a/\sqrt{2}-1}{\varepsilon}.$$

The feasible set of (5) is $[-\sqrt{2},\sqrt{2}]$, and since $a>\sqrt{2}$, its solution is $\theta=\sqrt{2}$, at which the constraint is active. The KKT condition $\sqrt{2}-a+\nu\varepsilon\sqrt{2}=0$ gives the multiplier $\nu=(a/\sqrt{2}-1)/\varepsilon>0$, which is the threshold above. □

# F Non-Identifiability of the Latent Prediction Error

For a predictor, consider its rollout map $R : (z_{\le t}, g') \mapsto (\hat{z}_{t+1}, \dots, \hat{z}_{t+H})$, and write $L(R)$ and $C(R)$ for (1) and (3) evaluated with this map. The JEPA loss evaluates $R$ only at pairs $(z_{\le t}, g)$ in which $g$ is the gravity of the real window that produced $z_{\le t}$; we denote the set of such pairs by $\mathcal{S}$.

**Proposition 2.** *Suppose the probe is affine, $h(z) = Wz + b$ with $W \in \mathbb{R}^{2K \times m}$, where $m > 2K$ is the latent dimension, let $u \in \ker W$ with $\|u\| = 1$, and assume that $\mathcal{S}$ is measurable. For a rollout map $R$ and $\lambda \in \mathbb{R}$, define*

$$R_\lambda(z_{\le t}, g') := R(z_{\le t}, g') + \lambda\, \mathbf{1}\{(z_{\le t}, g') \notin \mathcal{S}\}\,(u, \dots, u),$$

*and let $\hat{z}^\lambda_{t+s}$ denote its predictions. Then $L(R_\lambda) = L(R)$ and $C(R_\lambda) = C(R)$ for every $\lambda$. Moreover, for every distribution $\mathbb{Q}$ of $(z_{\le t}, g')$ and target latents $z_{t+1:t+H}$ with $\mathbb{Q}(\mathcal{S}^c) > 0$ and $\mathbb{E}_\mathbb{Q}\|\hat{z}_{t+s} - z_{t+s}\| < \infty$, we have $\mathbb{E}_\mathbb{Q}\|\hat{z}^\lambda_{t+s} - z_{t+s}\| \to \infty$ as $|\lambda| \to \infty$.*

*Proof.* On $\mathcal{S}$, $R_\lambda = R$, so the integrand of (1) is unchanged and $L(R_\lambda) = L(R)$. For any $(z_{\le t}, g')$, the decoded predicted positions satisfy $h(\hat{z}^\lambda_{t+s}) = h(\hat{z}_{t+s}) + \lambda \mathbf{1}\{\cdot\}\, Wu = h(\hat{z}_{t+s})$, while the decoded context positions are computed from real latents and do not depend on the rollout map. Hence every $a_{k,\tau}$, and therefore $c$, is unchanged pointwise, and $C(R_\lambda) = C(R)$. Finally, by the triangle inequality,

$$\mathbb{E}_\mathbb{Q}\|\hat{z}^\lambda_{t+s} - z_{t+s}\| \ge \mathbb{E}_\mathbb{Q}\Big[\mathbf{1}_{\mathcal{S}^c}\big(|\lambda| - \|\hat{z}_{t+s} - z_{t+s}\|\big)\Big] \ge |\lambda|\,\mathbb{Q}(\mathcal{S}^c) - \mathbb{E}_\mathbb{Q}\|\hat{z}_{t+s} - z_{t+s}\|,$$

which tends to infinity as $|\lambda| \to \infty$. □

*Remark* 14. A counterfactual pair $(z_{\le t}(w), g')$ with $g' \neq g$ lies outside $\mathcal{S}$ whenever the gravity of a window can be recovered from its context, e.g., from three or more free-fall frames by Lemma 1 when $E$ is injective. For a nonlinear probe, the same argument applies with perturbations along the level sets $\{z : h(z) = h(\hat{z}_{t+s})\}$, provided these level sets are unbounded. Proposition 2 concerns rollout maps; whether a given architecture realizes $R_\lambda$ is a separate question. A bound on the latent error requires additional structure, for example that latents of free-fall frames are a Lipschitz function of object positions and static scene content, and that predictions remain in the range of this function.